Natural Language Processing
Final Project Guide and Challenge Tracks

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Abstract

One question organises every project this term: *with today's AI tools, what LLM research can a team of two or three finish in four weeks and write up as a research report?* The final project is a small research study in natural language processing: choose one question, build a baseline, and test a targeted change under a fixed budget. The deliverables are a poster presentation, a research report, and reproducibility materials. This guide merges the project briefs and the Lecture 8 topics into eight tracks (A–H, 79 briefs), ordered from the most popular and easiest to grasp to the more specialised, with competitions and entrepreneurship last. It follows the course web page section by section, https://nlp-course-cuhksz.github.io/final-project.html, which adds embedded videos, a track finder and a Chinese version. This guide covers CSC5051, CSC6052, and MDS5110. Version of November 13, 2026.

1 The question, and eight tracks that answer it

The final project is a small research study in natural language processing. You choose one question, build a working baseline, and test **one targeted change** under a fixed budget. The deliverables are a poster presentation, a research report, and reproducibility materials. What is graded is not the size of the system but the clarity of the question, the fairness of the comparison, and the honesty of the write-up: a project that shows one clean success and one well-explained failure is a good project.

AI coding tools have changed what a four-week project can be. A team can now stand up a 3D scene, a game, a training run, a dataset pipeline or an agent harness in days instead of weeks. We therefore ask every team the same thing: *what can you learn in a month that you could not learn without doing the experiment?* The PDF guide's appendices and the Lecture 8 topics are merged into the **eight tracks** below, ordered from the most popular and easiest to understand to the more specialised, with competitions and entrepreneurship last. Tracks are entry points, not fences: a campus NPC project is Track C but borrows memory designs from Track A and evaluation discipline from Track D. Every brief carries tags for the compute it needs; the filter hides briefs you cannot run.

*Course instructor.

Track	Title	What you can build; briefs and compute
A	Agents & harness: tools, skills & iterative repair (App. A)	Harness engineering: pelican-on-a-bicycle in three.js, playable games, machine design with simulators, a reusable skill people actually install, an agent that writes and audits a survey, an MLE agent that enters Kaggle on its own. *11 briefs · API model often enough*
B	Train your own model: vertical models, new modalities & nanoGPT (App. B)	A nanoGPT speedrun experiment, your own twist on SFT or RL, the Tutorial 2 recipe applied to a vertical domain, adapters for speech, bio-signals and time series, weight-space programming, SAE interventions. *12 briefs · GPU needed*
C	HCI: interactive 3D/VR worlds, digital humans, education & interaction (App. C)	Campus with LLM NPCs, adaptive three.js scenes, dancing avatars, a visual personal assistant, a WebXR assistant, a JEPA-style world model; textbook skits, math videos, 3D lessons, an AI lecture series, content pipelines for special audiences, when to ask. *20 briefs in two parts · demo-heavy*
D	Evaluation, datasets & evidence (App. D)	Benchmarks the top models cannot pass yet, physician-style rubrics, judge audits, LLM-text detection, injection defenses; datasets per training stage, evidence-based RAG corpora, AI-built corpora with verifiers. *14 briefs in two parts · API or CPU*
E	AI for society: position papers, audits & interventions (App. E)	Does AI widen gender inequality, silence minority languages, exclude people with disabilities, or leave the Global South behind? Argue it like an ICML position paper and measure it. *6 briefs · no GPU*
F	Efficiency, on-device & GPU kernels (App. F)	Kernel generation with feedback, serving mixed workloads, quantization, speculative decoding, a phone benchmark, and an evolutionary (RSI-style) loop that optimises your own code. *9 briefs · one GPU or a phone*
G	Competitions (Chinese & international) (App. G)	Tianchi, Kaggle, NeurIPS 2026 and archived NLPCC/CCKS tasks, with the course protocol for turning a contest into a research project. *20+ contests · dates checked Nov. 3*
H	AI entrepreneurship: run a startup prototype (App. H)	AI tutor, AI seller, AI CEO, expertise training, text-to-game, AI + VR experiences, speech-based 3D printing, plus the business-model basics every AI founder should know. *7 briefs · users required*

New on November 9. An ML-engineering agent that enters a Kaggle competition on its own (A11); post-training variants of your own instead of a reproduction (B6); a visual personal assistant (C4); a clearer active embodied QA brief (C6); an AI lecture series with generated courseware, TTS narration and a recurring host (C20); multimodal long context with LongLLaVA (D1); bonus questions on identifying the model behind a relay API (D3), on aligning a draft model with different targets (F5) and on running a 2.78-trillion-parameter mixture of experts in 8 GB of RAM (F6); DeepTutor as a starting point for the AI tutor (H1).

Tags: **API** a hosted or small local model is enough · **1 GPU** a single 16–24 GB GPU for hours · **multi-GPU** several GPUs for days · **competition** has a matching live or archived contest. Posts from X cited as inspiration were located through search on Nov. 4, 2026 and could not all be opened directly; treat them as design prompts, not as verified results.

Use strong models, and say so. We encourage you to use the strongest model you can access as a planner, coder, judge or research assistant: an API model such as Claude Opus 5.5, or a strong open-weight model such as GLM-5.x or Qwen3-VL served locally. Log the model version and its role in every experiment. Where a model writes code or judges outputs, verify with executable tests, ground truth, or a human audit of a sample. For hosted competitions, use only the models and tools the organizer allows. The report must describe material uses of generative AI in coding, data, evaluation and writing.

1.1 The shape of a good four-week project

- **Week 1** · **Freeze:** task, held-out split, working baseline, primary metric
- **Week 2** · **Build:** the one change you are testing; smoke-test end to end
- **Week 3** · **Compare:** matched-budget runs, seeds, ablation, error analysis

- **Week 4** · **Reproduce:** re-run the key result from a clean clone; demo or result figure
- **Poster & report:** Dec. 17 / Dec. 25 question · result one limitation

Suggested research window: Nov. 13 → Dec. 11. Keep a smaller fallback experiment ready from week 1, so a failed main change still yields a result. Rule of thumb: if the baseline is not running by the end of week 1, shrink the task, not the comparison.

The four-week plan used by every brief on this page. Each track section fills in what "the one change" and "matched budget" mean there.

1.2 How to choose a topic (from Lecture 8)

Combine your expertise with AI. A previous major in mathematics, science, engineering, medicine, law or finance is an advantage, not a detour: AI for math, AI for materials, AI for the clinic you know.

Combine your background. If your native language or dialect is under-represented, train or audit a model for it. Local culture and heritage count: 黄梅戏, 甲骨文, 京剧, 中医, 相声.

A real need near you. A relative with diabetes who needs out-of-hospital management, a friend with a visual impairment, a family shop that needs a seller agent. Real needs make honest evaluations.

Your interest matters most. Four weeks is long enough to notice whether you care. Pick the question you would read about on a Sunday.

1.3 Where the briefs come from

Each track starts from published work that a team can extend in four weeks; every brief links to its papers, code and data, and the PDF guide lists them as references. Track A builds on executable agent benchmarks such as SWE-bench [1] and MLE-bench [2], and on agents that write surveys (AutoSurvey [3]). Track B asks for small, testable changes to post-training in the spirit of selective token training (Rho-1 [4]), reinforcement learning on high-entropy tokens [5] and simplified policy optimisation (DAPO [6], Dr. GRPO [7]). Track C draws on generative agents [8], egocentric assistants (EgoLife [9]), embodied question answering (OpenEQA [10]), and courses and lecture videos generated by AI (MAIC [11], Paper2Video [12]). Track D builds on rubric-based evaluation (HealthBench [13]), long multimodal context (LongLLaVA [14], MileBench [15]) and audits of which model an API really serves (model equality testing [16], Real Money, Fake Models [17]). Track E follows audits such as the dialect-prejudice study in Nature [18]; Track F starts from speculative decoding and the alignment of a draft model with its target (EAGLE-3 [19], DistillSpec [20]), and from engines that stream a trillion-parameter mixture of experts from disk on a single CPU (kimi-k3-in-c [21]); Track H from open tutoring systems (DeepTutor [22]) and the evidence that unguarded AI help can harm learning [23].

2 Schedule, deliverables and grading

Dates follow the course schedule (Shenzhen time, UTC+8). Submission links and presentation logistics are announced in class.

Date	Milestone
October 30	Final project released (with Lecture 6)
November 6	Lecture 7: LLM Agents and Harness Engineering (background for Track A)
November 13	Lecture 8: Final Projects and Research Sharing
November 20	**Team formation and optional one-page proposal due** (one week after Lecture 8)
December 17, 17:00	Poster submission due (the teaching team prints posters submitted on time; late posters are printed by the team)
December 18	Final project poster presentations: a 4-minute pitch per team, rated by other groups and TAs
December 25, 23:59	Final report due

The final project contributes 55 points to the course grade (poster 10, report 45). Bonuses: TA favourites (+2), instructor favourites (+1), early-bird submission (+2), code reproducibility (+1); a serious ethics concern costs 1 point; the project credit is capped at 55. Details are in the course grading policy.

2.1 Deliverables

Poster and presentation (10). State the question, baseline, main result and one limitation. A live demo or a recorded replay can illustrate the result. Use the sample poster template and look at the poster samples. Poster quality 1 · oral presentation 4 · overall assessment 5. Rated by other groups and TAs.

Final report (45). At most six pages of main text in the course report template: motivation, related work, method, evaluation, results, failures and limitations, plus a contribution statement and a generative-AI use statement. Read the writing tips first. Technical excitement 10 · technical soundness 15 · writing clarity 15 · individual contribution 5.

Reproducibility. Code on GitHub with a README, environment and model versions, evaluation inputs (or instructions for obtaining them), and the commands that reproduce the main result. Competition entries add the dated score or submission link. Reports are made public after the poster session unless you tell the teaching team. If you release the work later, please acknowledge the course.

2.2 Teams and topic selection

You may work alone or in a team; the suggested size is two or three. A larger team must explain how the workload and individual responsibilities justify it. Each student is assessed individually, and the report carries a contribution statement. If you collaborate with people outside the course, identify your own work.

The optional one-page proposal is worth writing whenever the research question, evaluation method or compute budget needs early feedback. State the question and motivation, the closest prior work, the data or environment, the baseline, the proposed change, the primary metric, the resource budget, and a smaller fallback experiment. A copyable template is in the research playbook.

Example contribution statement. Alice designed the evaluation and annotated the held-out cases. Bob implemented the baseline and the proposed method. Both ran experiments and wrote the report.

3 Research playbook

3.1 Experiment design

Before writing anything, read the writing tips page (from Lecture 8): a report should read like a film with no dull stretch. Begin with one working baseline and one measurable outcome. Compare your method on the same held-out tasks and report the variables that affect interpretation: model version, prompts, training tokens, time limits, hardware, API spend. Include at least one comparison that

isolates the main change, and show both a success and a failure with your explanation. A table of three numbers with seeds and a replay script is worth more than a page of screenshots.

- **One change.** Everything else fixed: model, prompts, data, budget, seeds, evaluation code.
- **Matched budget.** Same number of calls, tokens, simulator evaluations or training tokens per condition; wall-clock reported separately.
- **Held-out.** A split (or scenarios, rooms, prompts, briefs) that nobody tuned on; frozen in week 1.
- **Independent checker.** Executable tests, simulator state, ground truth, or a different model validated against a human-audited sample.
- **Seeds and variance.** At least two seeds for anything stochastic; report the spread.
- **Negative result kept.** If the change does not help, the error analysis is the contribution.
- **Replayable.** Traces, logs and configs in the repository; one command reproduces the headline number.

3.2 Proposal template

Optional, one page, due with team formation on Nov. 20. Submit through the channel announced by the teaching team.

> Title:
> Team (names, student IDs, course codes):
> Track and brief (e.g., Track A, brief A2) or custom:
>
> 1. Question and motivation (2–3 sentences). What will we know at the end that we do not know now?
> 2. Closest prior work (2–4 references) and what we do differently.
> 3. Data / environment / benchmark, and the held-out split we will freeze in week 1.
> 4. Baseline (what, from where, expected number).
> 5. The one change we test.
> 6. Primary metric (+ one cost metric: calls, tokens, GPU hours, latency).
> 7. Resource budget: compute (GPU type $\times$ hours), API spend, data access, accounts needed.
> 8. Four-week plan (W1 freeze, W2 build, W3 compare, W4 reproduce + demo).
> 9. Fallback experiment if the main change fails or the budget runs out.
> 10. Risks and what we need from the teaching team.

3.3 Report structure (six pages of main text)

Section	What reviewers look for
Abstract and introduction	The question in one sentence; the main result in one sentence; one limitation.
Related work	The three to six papers or systems you build on, and the gap you fill. Cite benchmarks and tools you used.
Method	Enough detail to reimplement: prompts, schemas, architectures, budgets, hyperparameters (appendix for the long version).
Evaluation	Data or environment, held-out split, metrics, checker, seeds, hardware, cost.
Results	The matched comparison, the ablation, the frontier or curve; variance; one figure a reader remembers.
Failures and limitations	A concrete failure case with your explanation; what the result does not show.
Contribution and AI-use statements	Who did what; where generative AI materially helped in coding, data, evaluation and writing.

3.4 Reproducibility checklist

- Public GitHub repository with a README that states the question and the headline number.
- Environment file and exact model versions (checkpoint hashes or API model IDs and dates).
- Evaluation inputs, or a script that downloads them, plus the frozen held-out split.
- The commands that reproduce the main table, and the replay script for any trace-based metric.

- For competitions: the dated score or submission link and the notebook or bundle that produced it.

3.5 How to present at the poster session

- Lead with the question and the comparison figure; the architecture diagram comes second.
- Bring a live demo or a recorded replay on a laptop; have the failure case ready to show, not only the success.
- Prepare a 90-second version and a 5-minute version of the story; posters are rated by other groups and TAs.

4 FAQ

Can I work alone? Yes. The suggested team size is two or three; a solo project should be scoped like one brief, not two.

Can we propose something that is not on this page? Yes. Use the proposal template, name the closest track, and make sure there is a baseline, a held-out split and one change to test. The optional proposal exists for exactly this case.

Do we need a GPU? Not for most briefs in Tracks A, C, D, E, G and H, which run on an API model plus a laptop (or a phone, for F8). Tracks B and F assume at least one GPU; Kaggle and Colab are enough for the small-scale version of every brief there.

Is a position paper (Track E) really a final project? Yes, if it contains a falsifiable claim and at least one measurement you ran: an audit, a replication on current models, or a small survey. A literature review alone is not enough; a survey written by an agent and audited by you (brief A10) is.

Can a startup prototype (Track H) be our project? Yes. Treat it as a product experiment: two variants on the same user tasks, a metric fixed in advance, real users, a cost table and a business memo. Keep user data minimal and consented.

Can we read the page in Chinese? Yes: the 中文 button at the top right switches the text (links and citations stay in English). Deliverables are written in English.

How much API spend is reasonable? Plan it in the proposal and report it in the paper. A matched-budget design keeps spend predictable; most projects fit in a modest budget because the held-out set is small and fixed.

Can we use AI coding agents to write the code? Yes, and we expect you to. Describe material uses in the report, keep humans responsible for the experimental design and the claims, and verify generated code with tests.

Can we continue our own thesis or lab research? Yes, if the four-week study is a clearly delimited piece with its own baseline and comparison, and you identify your own contribution versus prior work by you or collaborators outside the course.

Is a human user study required for Track C or E? No. Scripted evaluations with a fidelity check and a transfer task are enough. If you do run a study, keep it small, anonymous and free of sensitive data, get consent, and describe the protocol in the report.

The competition ends after Dec. 25. Can we still use it? Yes. Course deliverables follow the course schedule; report your local held-out results and add the public score when it exists.

What if our main change does not work? Report it. A clean negative result with an error analysis and a working fallback experiment is a complete project. What fails a project is an unfinished baseline or a comparison that was never matched.

Will our report be public? Reports are made public after the poster session unless you notify the teaching team. If you release the work later, please acknowledge the course.

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A Agents & harness: tools, skills & iterative repair

Lecture 7 framed an agent as *model + harness*. This track is about the harness: the loop that runs the model's output, observes what happened (compiler errors, failing tests, a rendered frame, a simulator measurement, a sealed grader), and feeds that evidence back. The question is almost always the same: **at a matched budget, does a loop that sees executable feedback beat one-shot generation or independent sampling, and what does each success cost?** Because the model is usually an API call, this track is the most accessible on the page; the engineering effort goes into isolation, replay and scoring. Three 2026 additions: **skills**, reusable instruction folders that any agent can load (brief A9); **agents that do research**, from writing a survey to running a sealed self-improvement loop (A10, A5); and an **ML-engineering agent** that enters a Kaggle competition on its own and is scored by the leaderboard (A11).

What "matched budget" means here. Fix the number of model calls, tokens, simulator evaluations or wall-clock time, and give the same budget to every condition. Report runtime separately. Run every artifact in a clean, isolated environment (a fresh container or browser context), and store the full generation trace so the comparison can be replayed.

A.1 Research questions this track asks

- Which kind of feedback matters: error text, test results, rendered evidence, or a profiler trace? Which one gives the largest gain per extra call?
- Does staged planning (plan, then build, then replay and repair) beat a direct build agent on executable realization, not just on code volume?
- Can an agent revise a mechanism more successfully when it sees measured constraint violations than when it only sees "failed"?
- Under a sealed evaluation, does iterative revision that keeps successful changes transfer to hidden data, or does it overfit the visible split?
- How much of the quality−latency−cost frontier is reachable with test-time scaling alone (self-consistency, verifiers, search) before training anything?
- When a tool call fails, what recovery policy (plain retry, typed error feedback, bounded repair) gives the best recovery rate per unit cost?
- Does a packaged skill change what an agent produces on held-out tasks, and does the gain survive a change of agent or model?
- When an agent writes a literature survey, which pipeline stage removes the most fabricated citations per unit of cost: retrieval, outlining, or a post-hoc checker?
- On a Kaggle task, does tree search over solution drafts beat a greedy debug-and-improve loop at the same GPU budget, and does the agent's local validation score predict the private leaderboard?

A.2 Featured briefs

A1. From one sentence to an interactive 3D world: a procedural pelican on a bicycle and a hand-built harness Adapt Simon Willison's "pelican riding a bicycle" drawing benchmark to a browser 3D coding task. The model must return a single runnable HTML file. Freeze the prompt:

> Create a polished, interactive three.js scene of a pelican riding a bicycle. Construct the pelican, bicycle, and environment with procedural 3D geometry. Animate the pedaling so the legs follow the pedals, the wheels rotate consistently with movement, and the body moves in coordination. Set up the scene, lights, camera, and interactive controls in code. Return one runnable HTML file. Use no SVG, 2D canvas drawing, external models, or prerendered animation.

Then build the harness: run each artifact in an isolated browser, record console errors, capture fixed viewpoints and pedal phases, and replay a camera interaction. The harness is the research contribution; the pelican is the test case.

Compare. one-shot generation · a repair loop that receives browser errors and rendered evidence, at a matched call budget; a few held-out prompt variants (another animal, another vehicle).
Measure. launch success, visual recognizability, pedal−leg−wheel coordination, working controls:

automated checks where possible, a short blinded rubric for visual criteria.
Deliver. the harness, all generated HTML files, screenshots per phase, rubric sheets, and a results table across models.
Week 1: freeze prompt and rubric; Week 2: harness: browser isolation, capture, replay; Week 3: matched-budget trials, held-out variants; Week 4: results, reproducible artifacts. Start from: the original SVG prompt and results · Willison's 2025 talk and 2026 update · three.js docs · Playwright for isolated rendering and screenshots.

A2. Building playable games with an AI workflow on GameCraft-Bench [our lab] GameCraft-Bench (FreedomIntelligence) asks whether a coding agent can turn a natural-language game spec into a complete, playable 2D Godot game: 140 tasks across 15 families (platformer, strategy, tycoon, roguelike, ...). A game counts only if scripts, scenes, assets and input all work together; a verifier launches the project, replays submitted input traces, and a rubric plus multimodal judge scores core mechanics, content depth, functional visuals and presentation. Wrapper scripts exist for Claude Code, Codex and Kimi Code.

Compare. a direct build agent · plan-then-build · plan-build-replay-repair, on a fixed subset spanning two or three families, at a matched token or time budget.
Measure. project launch rate, implemented mechanics, held-out replay completion, cost per playable project.
Deliver. launchable projects, replays, a failure taxonomy, and the harness configuration.
Week 1: install, reproduce one task; Week 2: implement the three agent variants; Week 3: run the fixed subset; Week 4: replays, failure analysis. Start from: GameCraft-Bench repository · website with demos and leaderboard · paper [24] · Godot 4.

A3. AI designs a machine and iterates against a simulator Give an agent a parameterized gear pair or linkage and a brief: a motion ratio, a clearance, a size limit. Let it propose dimensions, run a simulator or an independent kinematic checker, and revise from the measurements. Keep the whole project to one mechanism family, one checker, and one replay that shows the initial failure and the revised design.

Compare. one-shot design · independent designs at the same generation budget · a parameter sweep · the revise-from-measurements loop; match simulator evaluations, report runtime separately.
Measure. feasible-design rate, objective error, calls or time to first feasible design on unseen briefs.
Deliver. the checker, all design traces, and an interactive rendering of the iteration.
Start from: demo inspiration · Airsup Machine Skill (one-line machine description → interactive three.js simulation) · Physics-in-the-Loop (IJCAI 2026) · CadQuery / build123d for parametric geometry · MuJoCo or Genesis for simulation · BuildArena S02 if your prototype is ready by Nov. 14.

A4. A minimal agent with a recovery mechanism (adapted from Stanford CS329Z) CS329Z's first homework builds a company assistant with no agent framework: a chat-completion call and code you write yourself. Do the same with one or two tools, then evaluate on tasks with controlled transient failures (timeouts, rate limits) and invalid arguments.

Compare. plain retry · typed error feedback · a bounded repair policy, at the same tool-call and model budget.
Measure. recovery rate, cost per successful task, side effects (unwanted writes, duplicate actions).
Deliver. the harness, replay logs, and a table of failure classes.
Start from: CS329Z: Engineering AI Agents (lecture 6, design patterns and scaffolds) · Building effective agents · MCP for tool definitions · DSPy once the plain version works.

A5. Improve a working method under a sealed evaluation (RSI-Exam) RSI-Exam gives an agent a task, a visible development split, and an "autoresearch protocol": loop through experiments, record every version, aim for what transfers. The artifact (a method, or a harness driving a frozen model) is scored once on hidden data in a sealed verifier container. Use one public task with a grader, or define a small task of your own.

Compare. independent attempts · iterative revision that retains successful changes, under a resource cap; run each frozen artifact in a clean environment.
Measure. development vs. held-out gain (the overfitting gap), cost, and at least one failure trace.
Deliver. the task bundle, a repeatable comparison, and the revision history.

Start from: rsi-exam.ai (88 tasks, six domains, leaderboard) · GitHub · task guidelines and contribution process (the organizers list Dec. 15, 2026 for RSI-Exam 2.0 task contributions and review proposals before implementation).

A6. Test-time scaling: the quality–latency–cost frontier Improve solution quality by spending more compute at inference, at a stated latency and cost budget. Methods: self-consistency over diverse samples, reflection loops, solution + verifier dual tracks, beam or tree search, evidence sampling with self-checking. Tasks: grade-school to olympiad math, code generation or repair, knowledge QA with citations.

Compare. a sweep over "thinking steps" and "verifier strength" for two or three methods on the same problems.
Measure. accuracy or Pass@k against latency and token cost; robustness across seeds.
Deliver. a minimal reproducible harness and the frontier plot.
Start from: Let's Verify Step by Step [25] · outcome-supervised value models [26] · the public AIMO 3 problems as a hard math set · Lecture 6 slides.

A7. A repository-repair agent on real issues Agents repair real repository issues and held-out tests score the patches. The Gemma 4 Developer Agent competition on Kaggle (entry Nov. 25, final Dec. 2) uses an offline scoring harness with an approved Gemma 4 variant (31B, 12B, E4B or E2B). Even without entering, SWE-bench-style tasks make a clean course project.

Compare. repository search alone · search plus a test-and-repair loop, on the same issues and budget.
Measure. pass rate on held-out tests, tool calls, runtime, cost per solved issue.
Deliver. the agent, trajectories, and a per-issue failure analysis.
Start from: Gemma 4 Developer Agent · SWE-agent · SWE-bench · Terminal-Bench.

A8. Program search vs. direct prediction on ARC, and memory in interactive ARC-AGI-3 The 2026 ARC Prize prediction tracks closed on Nov. 2, but the public tasks remain. Compare direct grid prediction with executable program search on ARC-AGI-2, or test how an agent's memory design affects progress in the public ARC-AGI-3 interactive environments. Allocate the search budget across tasks and report it.

Compare. direct prediction · program search · search with a verifier, at a matched call budget; or flat vs. summarised vs. retrieval memory in ARC-AGI-3.
Measure. tasks solved, calls per solved task, failure categories.
Deliver. the harness and per-task logs.
Start from: ARC Prize guide · ARC-AGI-2 · ARC-AGI-3.

A9. Build a skill that people actually install, and prove it helps Since Anthropic published Agent Skills as an open standard (Dec. 2025), a "skill" is a folder with a short description, instructions and optional scripts that an agent loads only when a task matches. The same file works in Claude Code, Codex, Cursor and others, and directories such as skills.sh rank installs. Lecture 8 showed frontend-slides as an example; Claude-Code-Game-Studios (49 agents, 74 skills, 30k+ stars) shows how far the idea can go. Pick one recurring task your classmates do badly by hand (slides from a paper, a poster from a report, a literature table, data cards, a Godot scene, a Manim lesson), write the skill, and treat it as a research object: does the skill change what the agent produces, and for whom?

Compare. the bare agent · skill v1 · skill v2 after eval-driven revision, on 20–40 held-out task prompts; same model, same budget. Bonus condition: the skill in a second agent (Codex or Cursor).
Measure. task success on a blinded rubric, tokens and wall-clock per task, failures by type; the skill's own token cost at load; installs, stars or classmates' repeat use as a secondary signal.
Deliver. the skill repository (SKILL.md, scripts, examples), the eval suite, the three-condition table, and a short "when not to use it" section.
Week 1: task, 30 prompts, rubric, bare-agent baseline; Week 2: skill v1; run evals; Week 3: revise to v2; second agent; blinded rating; Week 4: publish, write-up. Start from: Agent Skills specification · anthropics/skills (document, design and MCP-builder skills) · superpowers · vercel-labs/agent-skills · huggingface/skills · `claude plugin eval` for measuring whether a skill changes outputs · inspiration posts on X: 10 skills that turn Claude Code into a team, the skill-eval playbook, video-studio skills, slide-creation skills.

A10. An agent that writes the related-work survey, audited against a human one Lecture 8 listed "a survey of a specific LLM topic" as a project; in 2026 the interesting version is to let an agent write it and to measure what it gets wrong. AutoSurvey (NeurIPS 2024) and the AI Scientist (Sakana, whose paper passed workshop peer review in 2025) are the references. Choose a narrow topic with an existing human-written survey of 100–300 references, hide that survey, and compare a one-shot prompt with a pipeline: retrieval from arXiv and Semantic Scholar, an outline step, section writers, and a citation checker that verifies every reference exists and says what the text claims.

Compare. one-shot survey prompt · pipeline without the checker · pipeline with the checker, same topic, same model, same token budget.
Measure. fabricated or mis-attributed references, coverage of the hidden survey's reference list, taxonomy agreement with the human survey (blinded rubric), cost.
Deliver. the pipeline, three generated surveys, the audit sheets, and a list of claims the agent made that no cited paper supports.
Start from: AutoSurvey [3] (NeurIPS 2024) · The AI Scientist's first peer-reviewed publication · Semantic Scholar API · OWL and OpenManus for autonomous agent scaffolds · pairs with D4 (faithfulness of AI-written scientific statements).

A11. An MLE agent that enters a Kaggle competition on its own Machine-learning engineering is the agent task with the cleanest outside score: a Kaggle leaderboard. MLE-bench (OpenAI) packages 75 past Kaggle competitions with their medal thresholds and a local grader, and by 2026 the best open agents (MLEvolve, ML-Master 2.0, AIDE-style tree search) earn a medal on about three quarters of its 22-task Lite split. Build an agent that reads a competition page and its data, plans, writes and debugs training code, validates locally and submits through the Kaggle API. Develop it on MLE-bench, then point it at a *live* competition (a monthly Playground Series episode or a Getting Started competition) whose private leaderboard nobody has seen.

Compare. a one-shot pipeline (one plan, one script) · a greedy debug-and-improve loop · tree search over solution drafts (AIDE style), with the same GPU hours and model calls; then the best variant with and without a memory of earlier competitions.
Measure. valid-submission rate, medal or percentile on 5–10 MLE-bench Lite tasks, the gap between local cross-validation and the leaderboard (does the agent overfit?), GPU hours and API cost per medal, and the dated rank on the live competition.
Deliver. the agent, every solution tree and log, a per-competition table, and the live submissions with timestamps and scores.
Week 1: MLE-bench installed; two Lite tasks solved by hand and graded; Week 2: one-shot and greedy agents; Week 3: tree search, memory, matched-budget runs; Week 4: live Kaggle entry, leaderboard-gap analysis. Start from: MLE-bench paper [2] and repository with leaderboard · AIDE [27] (aideml) · ML-Master [28] · MLEvolve · R&D-Agent (Microsoft) · AutoKaggle [29] · Kaggle API (`kaggle competitions submit -c -f submission.csv -m "..."`) · Kaggle competitions. Read the rules first: daily submission limits, team and external-data rules, and any policy on automated submissions apply to the agent exactly as they apply to you.

A.3 Watch, read, play

- **Course material** Lecture 7: LLM Agents and Harness Engineering · Effective harnesses for long-running agents · Harness engineering (OpenAI) · Harness engineering for coding-agent users
- **Stanford CS329Z: Engineering AI Agents** (Diyi Yang, Michael Ryan, John Yang; Fall 2026) syllabus, slides and homework · Chinese study site with translated readings Lecture 6 "Agent design patterns and scaffolds" is the backbone reading for Track A.
- **Simon Willison's pelican benchmark** The last six months in LLMs, illustrated by pelicans on bicycles (video + annotated slides) · 2026 in LLMs (so far) His rule: when models draw a perfect pelican, change the animal and the vehicle. Your held-out variants do the same.
- **GameCraft-Bench demos** website with playable trials and the leaderboard of harness/model configurations Watch how the verifier replays input traces before you design your own replay-based scoring.
- **Agent benchmarks from Tsinghua / Z.ai** AgentBench and VisualAgentBench · Open-AutoGLM (open phone-agent model and framework) · slime (RL framework used for agentic post-training)
- **Inspiration posts (X)** give a coding agent a strong reference, the live URL and the source repo (Meng To) · 25 open-source repos for vibe-coding games, 10+ agents in parallel · the 2026 Vibe Coding Game Jam rules (90% AI-written code) · /simplify and /batch as skills in Claude Code · "why isn't the nanoGPT speedrun fully AI-automated by now?" Design prompts only; the posts were located through search and not independently verified.

- **Agent workflows and skills** Agent Skills standard · anthropics/skills · skills.sh leaderboard · Dify (workflow builder) · OpenManus · OWL · MCP and MCP servers From the Lecture 8 "agent workflow" slide: Dify workflow, RAG corpora, MCP, tool hubs; autonomous agents OpenManus and OWL.
- **Tools for the harness** Claude Code · Codex CLI · OpenClaw · LangGraph · DSPy · MCP servers · OSWorld for computer-use tasks

A.4 More research questions in this area

- **Full replanning vs. local plan repair** after a failed step: which keeps task success higher at the same call budget?
- **Robustness to DOM or UI changes and prompt injection** for web agents: fix a task set, perturb the pages, measure success and side-effect rate (overlaps with D3).
- **Memory compression** for long-running agents: at what context size does summarisation start costing task success?
- **Multi-agent vs. single-agent** on the same tasks and total budget: does the division of labour pay for its coordination cost?
- **Hospital-as-environment agents.** Our lab's Agentic Hospital models a hospital as an actionable environment (mocked HIS, role simulation, governance); a bounded clinical or operational task with its rubric makes a Track A or D project.

Pitfalls we see every year. The repair loop sees the held-out tests. The judge model wrote the code it judges. Budgets differ between conditions ("the loop just ran longer"). No isolation, so flaky runs are counted as failures. Reporting success rate without cost per success. Store traces; a reviewer should be able to replay any number in your table.

B Train your own model: vertical models, new modalities, nanoGPT experiments & internals

This is the track for teams who want to touch weights. The philosophy is borrowed from Stanford CS336 and from Tang Jie's Advanced Machine Learning course at Tsinghua: **to understand it, build it**. A 0.1B–0.5B language model trained from random weights is now a one-GPU, one-week exercise, and post-training with SFT, DPO or GRPO fits in a Kaggle session (see Tutorial 2). Two newer directions make the track more interesting than "train a small GPT": *weight-space programming* (a frontier model reading and editing a checkpoint directly) and *mechanistic interpretability at scale* (do sparse-autoencoder features have consistent causal effects?). Whatever you choose, change one thing at a fixed budget and plot the learning curve. Three briefs added this term follow Lecture 8 directly: a **nanoGPT speedrun** experiment in architecture engineering (B10), a **vertical model** built with the Tutorial 2 recipe (B11), and a **new modality** such as speech, bio-signals or time series (B12). For post-training, B6 asks for a **variant of your own** (which tokens learn, which data, a cheaper PPO), not a reproduction.

Compute reality check. A 0.1B model on 1–2B tokens trains in hours on one 24 GB GPU; 0.5B on a Chinchilla-style token budget needs several GPU-days. Establish throughput (tokens per second, peak memory) on the small model first, then decide whether the 0.5B run is affordable. Free tiers: Kaggle (about 30 GPU hours per week, T4/P100), Colab; paid by the hour: AutoDL and cloud providers. A report with a complete small-scale result beats an unfinished large one.

B.1 Research questions this track asks

- At a fixed token budget, what does one data intervention (deduplication, quality filtering, a code-to-text mixture change) do to held-out loss and a downstream task?
- Two model sizes at a matched approximate compute budget: which wins, and does the answer change with the data mixture?
- Can a frontier model repair a specified failure by editing a checkpoint directly, with a bounded edit budget and no gradients? How does the patch compare with a random edit of the same size and norm?
- Do SAE features that look interpretable on examples have a consistent causal effect across prompt templates, compared with random features and a neuron baseline?
- Which small change to SFT or RL (which tokens receive gradient, which problems are sampled, how the loss is normalised, how much of PPO can be dropped) buys the most accuracy per GPU hour, and does the gain survive at large Pass@k?
- How much does a domain-adapted model forget, and does a small replay mixture fix it?
- How should a fixed LoRA parameter budget be allocated across layers?
- At matched wall-clock on your GPU, does one architecture or optimizer change from the nanoGPT speedrun changelog still help, and does the gain survive a change of scale?
- For a vertical domain, which post-training stage (continued pre-training, SFT, preference optimization, retrieval) buys the most at a fixed GPU budget, and are retrieval and training substitutes or complements?
- How should time-series or bio-signal data enter a frozen LLM: serialised text, a soft-prompt adapter or cross-attention, and which is most data-efficient on held-out subjects?

B.2 Featured briefs

B1. From scratch to a 0.5B training experiment Under a fixed token budget, compare random sampling with a single data intervention (deduplication, an educational-quality filter, a different code fraction), keeping architecture, tokenizer and optimizer fixed. Start with a small randomly initialized model to establish the learning curves, then run both conditions at 0.5B only if measured throughput and available compute permit.

Compare. baseline data vs. one intervention at the same token count, two seeds at the small scale.
Measure. held-out loss curves, one downstream task, measured tokens per second and peak memory.
Deliver. configurations, matched checkpoints or logs, and a scaling note on what 0.5B would cost.
Week 1: tokenizer, model, data loader; throughput; Week 2: baseline run + intervention run (small); Week 3: seeds, downstream eval, optional larger run; Week 4: curves, cost table, write-up. Start

from: Chinchilla [30] · nanochat (full pipeline, small budget) · CS336 assignment repos · FineWeb-Edu · datatrove for dedup and filtering.

B2. Train and profile a small language model from scratch (adapted from Stanford CS336) Follow the CS336 sequence: tokenizer, decoder, optimizer, training loop, all written by you. Then run a clean experiment: two model sizes at a matched approximate compute budget, or one throughput setting (sequence length, batch size, attention kernel, mixed precision) varied with the model held fixed.

Compare. two sizes at matched compute, or one throughput knob.
Measure. held-out loss, actual tokens per second, peak memory, and the measured budget for any optional 0.5B run.
Deliver. configurations, logs, a profiling summary.
Start from: CS336: Language Modeling from Scratch (Spring 2026 lectures on YouTube; Chinese subtitles on Bilibili, see below) · torch.profiler.

B3. Reproduce a model result and test one data change (adapted from Tsinghua AML (Tang Jie)) The Tsinghua course pairs implementation exercises (reproducing GLM-4-9B basics, fine-tuning plus RAG, visual LLMs, advanced reasoning) with a team research project. Reproduce a small published result with its configuration, then compare the original data with one deduplicated or quality-filtered variant. Keep tokenizer, architecture and token budget fixed.

Compare. original data vs. one filtered variant at the same budget.
Measure. matched learning curves and a held-out comparison.
Deliver. the run configuration, curves, and a data card for the filtered set.
Start from: AML 2024 course archive (homework and project directions) · GLM-5 repo · TRL.

B4. Weight-space programming: a language model reads and edits a checkpoint Takehi, Teoh, Zhang and Khattab (MIT CSAIL, Nov. 2026) report that a frontier model can infer behaviour from raw weights and edit them to change it: identifying which game a PPO policy plays from its weights, fixing a Kinetix car policy by changing 3.5% of weights, and lifting Qwen3.5-0.8B from 7.9% to 48.1% on MATH-500 by editing 0.016% of its weights with 50 forward passes. The same method gave no gain on Gemma 3 270M. Replicate the idea at a scale you control.

Compare. the proposed patch vs. the original checkpoint vs. a random edit matched for parameter count and norm; add a simple task-specific repair baseline when one exists.
Measure. held-out task success, retention on unaffected cases, edit size, across seeds; log every forward pass used during editing.
Deliver. one reproducible patch experiment on one model family (a tiny MLP or policy first; a small LM checkpoint is a stretch goal), with the exact tensor diff.
Week 1: tiny model with a reproducible failure; edit interface; Week 2: editing protocol with budget and dev set; Week 3: baselines, seeds, retention; Week 4: stretch: small LM; write-up. Start from: Sparks of Weight-Space Programming · full logs, code and datasets · live demo · edited Qwen3.5-0.8B.

B5. Mechanistic interpretability at scale: do SAE features have consistent causal effects? Use released Gemma Scope sparse autoencoders to study one behaviour, such as copying a repeated sequence. Select features from activation examples on development prompts, then intervene on held-out prompts. The question is whether an explanation that looks right on examples survives a causal test across templates.

Compare. feature intervention vs. no-op vs. random features at matched magnitude vs. a simple neuron baseline.
Measure. change in target-token log odds, effects on unrelated prompts, results by template; two SAE widths at comparable sparsity only after the pipeline works.
Deliver. the intervention pipeline, feature lists, and per-template results.
Start from: Gemma Scope · Scaling and evaluating sparse autoencoders · SAELens · TransformerLens · the AIMO Interpretability Challenge (NeurIPS 2026) for a reasoning-model variant.

B6. A twist on SFT or RL: train on the right tokens, the right data, or with a cheaper PPO This is not a reproduction. Pick one ingredient of SFT or RL, propose a variant of your own, and

show at matched compute whether it helps and why. Several of the most cited post-training results of 2024–2026 are changes of a few lines (which tokens receive gradient, which problems are sampled, how the loss is normalised), so the format fits four weeks and one GPU. Choose one family, read its starting points, and make your own twist the research question.

- **Which tokens learn**. *Published starting points:* Rho-1 [4] (selective language modelling: train only on the tokens a reference model marks as useful); high-entropy minority tokens [5] (RL gradients on the 20% "forking" tokens match or beat full updates); DFT [31] (rescale each SFT token loss by its probability, one line); *Twists worth testing (examples):* select tokens by disagreement between two checkpoints; apply the high-entropy mask to SFT instead of RL; stop gradient on tokens copied from the prompt; select by the verifier's credit for each step
- **Which data**. *Published starting points:* LIMO [32] and s1 [33] (about 1k curated examples); 1-shot RLVR [34] (RL on a single training problem); dynamic sampling in DAPO [6] (drop groups whose rewards are all equal); *Twists worth testing (examples):* a curriculum by the current policy's pass rate; deduplication by solution strategy rather than by text; replay of already-solved problems to limit forgetting; rewriting hard problems into easier stepping stones
- **How the loss is computed**. *Published starting points:* Dr. GRPO [7] (remove the length and standard-deviation normalisation); DAPO (clip-higher, token-level loss, overlong shaping); GSPO [35] (sequence-level ratios); Clip-Cov / KL-Cov [36] against entropy collapse; *Twists worth testing (examples):* a coefficient that holds policy entropy at a target; clipping that depends on token entropy; a length penalty scaled by problem difficulty; mixing SFT and RL losses on the same batch
- **Making PPO cheaper**. *Published starting points:* ReMax [37] and RLOO [38] (no value model); GRPO's group baseline (DeepSeekMath [39]); asynchronous rollouts (AReaL [40]); LoRA for the policy; *Twists worth testing (examples):* reuse each rollout batch for two updates with importance correction; stop rollouts early once the verifier can already decide; skip prompts the policy always solves; report accuracy per GPU hour and peak memory, not accuracy alone

Compare. the unmodified baseline recipe · the published variant · your twist, with the same base model (for example Qwen3-0.6B or 1.7B), the same data, GPU hours and seeds; plus one ablation that removes the part of your twist you believe matters.
Measure. accuracy and Pass@k on MATH-500, GSM8K or a code set (does the gain survive at large k?), an out-of-distribution set, entropy and response length during training, GPU hours and peak memory per point gained.
Deliver. the diff against the baseline trainer (ideally a few dozen lines), configs, seeds, learning and entropy curves, and one paragraph that explains *why* the change works or fails.
Week 1: baseline recipe running end to end; Week 2: published variant reproduced at small scale; Week 3: your twist and its ablation, two seeds; Week 4: Pass@k, out-of-distribution set, write-up. Start from: Tutorial 2 notebooks (SFT, reward model + PPO, GRPO, DPO) · TRL · verl · OpenRLHF · slime · SimpleRL · for the Pass@k question: does RL expand reasoning beyond the base model? [41] · SFT memorizes, RL generalizes [42] · Lecture 6 slides.

B7. Domain or low-resource language adaptation, with forgetting measured Adapt a small model to a domain (medicine, law, a dialect) with a small curated corpus. The point is not the domain score alone but what is lost, and whether a replay mixture recovers it.

Compare. base · domain-adapted · domain-adapted with 10–20% replay of general data.
Measure. domain task gain, general-task regression, calibration.
Deliver. the recipe, data card, and a forgetting table.
Domain examples from our lab: HuatuoGPT [43], HuatuoGPT-II [44] · Lecture 9: Medical LLMs · PEFT.

B8. Process rewards for stepwise reasoning Train or prompt a step-level verifier and use it to re-rank or guide a small reasoning model. Compare outcome-only reward with process reward at the same number of labelled or generated examples.

Compare. outcome reward · process reward · both, same budget.
Measure. accuracy, verifier precision on held-out steps, cost per correct answer.
Deliver. the verifier, labelled steps, and the comparison.
Start from: Let's Verify Step by Step [25] · DeepSeekMath (GRPO) [39].

B9. Where should a fixed LoRA budget go? Given a fixed number of trainable parameters, compare uniform LoRA ranks with allocations that favour attention, MLP, or late layers; include LoRA vs. QLoRA vs. adapters under one evaluation protocol, as in the Tsinghua AML week-14 assignment.

Compare. allocation schemes at equal parameter count.
Measure. task accuracy, memory, training time, variance across seeds.
Deliver. configs and a results table.
Start from: PEFT · Prefix-Tuning [45] · Tutorial 2 (LoRA/QLoRA).

B10. A nanoGPT speedrun experiment: one architecture or optimizer change at a fixed budget Keller Jordan's modded-nanogpt turned "train GPT-2 124M to 3.28 validation loss on FineWeb" into a public speedrun: 45 minutes in 2024, under two minutes by the end of 2025, about 40 seconds by August 2026, and a disputed 6.7-second entry in late October 2026 that uses n-gram hints built on idle CPUs. Every record is a documented change: the Muon optimizer, QK-norm, ReLU^2 MLPs, U-net skip connections, value embeddings, sliding-window FlexAttention, logit soft-capping, paired-head attention, fused Triton kernels. That changelog is a syllabus for architecture engineering. Pick **one** idea (or a new one from Lecture 4: MLA, gated linear attention, a hybrid Mamba block, nGPT normalisation, μP scaling), implement it in a single-GPU scaled-down version of the run, and measure it properly.

Compare. the current record script (scaled to your GPU and token budget) · the same script with your one change, at matched tokens and matched wall-clock, two seeds each.
Measure. validation loss against tokens and against seconds, tokens per second, peak memory, and one downstream probe (HellaSwag or a small QA set); report whether the change helps at equal time, not only at equal steps.
Deliver. a pull-request-style diff, both learning curves, the hardware and timing protocol, and a note on whether the result should transfer to the 8×H100 setting.
Week 1: reproduce the record at small scale; throughput; Week 2: implement the change; Week 3: matched-time runs, seeds, probe; Week 4: curves, diff, write-up. Start from: modded-nanogpt (records and changelog) · nanoGPT and nanochat · Muon · Epoch AI record timeline · the n-gram-hint record and the rules debate · agents running the speedrun autonomously · records on X: 39.9 s → 6.7 s explained, paired-head attention, a fused ReLU^2 Triton kernel, the medium track · Lecture 4: Transformer and beyond.

B11. A vertical model with the Tutorial 2 recipe: continued pre-training, SFT, preference optimization, RAG Lecture 8's first topic is the vertical LLM: medicine, law, finance, a trade, a university's own regulations. The four tools are continued pre-training on domain text, supervised fine-tuning on domain tasks, preference optimization (DPO or GRPO with a verifier), and retrieval. Tutorial 2 gives you the SFT, reward-model, PPO, GRPO and DPO notebooks on a 4-bit base with LoRA adapters; HuatuoGPT and HuatuoGPT-II are the lab's worked examples. The question is not "can we build one" but **which stage buys what** for your domain at a fixed budget.

Compare. base + RAG · SFT · CPT + SFT · SFT + DPO/GRPO, same base model, same held-out domain test, same GPU hours; add RAG on top of the best to see whether retrieval and training are substitutes or complements.
Measure. domain accuracy or rubric score, general-task regression (forgetting), hallucination rate on unanswerable questions, cost per stage.
Deliver. the recipe with configs, the domain test set with a data card, and a stage-by-stage table.
Week 1: domain data, test set, base + RAG baseline; Week 2: SFT, CPT runs; Week 3: DPO/GRPO, forgetting checks; Week 4: table, write-up. Start from: Tutorial 2 notebooks · HuatuoGPT [43], HuatuoGPT-II [44] (one-stage domain adaptation) · ORLM [46] (optimization modelling), GrammarGPT [47], RAG-Instruct [48] (other lab examples from Lecture 8) · TRL · LLaMA-Factory · Lecture 9.

B12. Teach an LLM a new modality: speech, bio-signals or time series through a small adapter
Lecture 8 (Topic 8) and Lecture 10 describe the recipe: a modality encoder, a small projector or cross-attention adapter, and an LLM that stays mostly frozen. The lab's Soundwave reached Qwen2-Audio quality on speech tasks with one-fiftieth of the data by adding a CTC alignment loss and better data engineering; OpenTSLM (Stanford, 2025) made 1B models reason over ECG, sleep and activity time series, beating GPT-4o on those tasks. The project: pick one signal type and one task family (speech commands, ECG rhythm QA, EEG sleep staging, accelerometer activity, sensor logs, stock or weather series) and ask **how time should enter the model**.

Compare. text-only baseline (numbers serialised as text) · soft-prompt adapter · cross-attention adapter, same frozen LLM and same training examples; ablate the alignment loss or the patch size.
Measure. task accuracy or F1 on held-out subjects (not held-out windows), data efficiency (curve over training-set size), and whether the model still answers text questions (retention).
Deliver. the adapter code, the subject-split evaluation, and the data-efficiency plot.
Week 1: signal, task, subject split, text baseline; Week 2: encoder + adapter v1; Week 3: second adapter, data-efficiency curve; Week 4: retention check, write-up. Start from: Soundwave [49] (code, our lab) · OpenTSLM [50] (checkpoints) · Time-LLM (ICLR 2024) · Chronos · ECG-FM · PhysioNet and MIMIC data · posts on X: an EEG foundation model on a $50 device, ECG-FM in JAMA Network Open · Lecture 10: Multimodal LLMs.

B.3 Watch, read, play

- Video: **Stanford CS336 (Spring 2026), Lecture 1: Overview and Tokenization**. Chinese subtitles: Bilibili (中英字幕); notes: cs336-blog.
- Video: **Stanford CS25 V6: The Ultra-Scale Talk**, scaling training to thousands of GPUs. Watch before claiming a multi-GPU run.
- **Tsinghua 高级机器学习 (Advanced Machine Learning), Tang Jie** 2024 archive (HW1 reproduce GLM-4-9B basics; HW2 fine-tuning + RAG; HW3 visual LLMs; HW4 advanced reasoning; projects in reasoning, multimodal, recommendation, education, science, healthcare, business) · Fall 2026 edition (news): tokenizer and Transformer from scratch, a 0.1B model end to end, a hand-written Triton attention kernel, fitting a scaling law, LoRA/QLoRA/adapter comparison, and a long-horizon self-evaluating agent as the final deliverable. Several of those assignments are briefs here: B1, B2, B9, F1, A5.
- **nanoGPT speedrun (X)** the 6.7-second record and the rules debate · 39.9 s: a 65B-entry n-gram table, new softmax and optimizer · 75.4 s from a faster ReLU^2 kernel · paired-head attention: 3% for zero parameters · why is this not fully automated yet? Each record is one documented change at a fixed target: the discipline brief B10 asks for.
- **Bio-signals and time series (X)** an EEG foundation model paired with a $50 device · ECG-FM, 1.3M ECGs, JAMA Network Open · NeuroLM: bridging language and EEG · a one-lead wearable ECG foundation model Evaluate on held-out subjects; window-level splits leak.
- **Weight-space programming** blog post · live demo · logs and code Read the negative result on Gemma 3 270M as carefully as the positive one on Qwen3.5-0.8B.
- **Training stacks** nanochat · CS336 assignments · TRL · PEFT · verl · OpenRLHF · slime · lm-evaluation-harness
- **Course material** Lecture 5: LLMs · Lecture 6: RL for post-training · Tutorial 2: SFT, LoRA/QLoRA, PPO, GRPO, DPO

Pitfalls we see every year. Two things changed at once. Conditions compared at different token counts. A benchmark that overlaps the training data. One seed. "We trained a 0.5B model" with no throughput or cost numbers. A learning curve that stops before the comparison is meaningful. Fix the budget, change one thing, show the curve.

C HCI: interactive 3D/VR worlds, digital humans, education & human−AI interaction

This track is about **interaction**: what changes when a language model is placed where people can see, touch, walk through or learn from it. Part 1 puts the model inside a 3D scene, a game, a headset, a robot simulator or an animated character and asks what it contributes (spatial language, memory, grounded action, real-time decisions). Part 2 takes a concrete piece of teaching material and builds something a learner can watch, rotate or experiment with, and asks when an assistant should ask, remember or stay quiet. Both parts end in a demo, and both are graded on the controlled comparison behind it: fix the scene or the lesson, hold out tasks, score with an independent checker, and measure interaction as task success, fidelity and transfer rather than "engagement".

Part 1 · Interactive 3D worlds, AR/VR, embodied AI & digital humans (briefs C1−C10)

Language models are leaving the chat box. This track puts an LLM inside a 3D scene, a game, a robot simulator or an animated character, and asks what the language model actually contributes: **spatial language, memory, grounded action, or interaction**. Every project here ends in something you can play in a browser or watch as a replay, but the report is graded on the controlled comparison behind the demo. The 2026 toolchain makes the demo part cheap: image-to-3D models export glTF assets that three.js loads directly, open world models produce editable scenes, and a frontier coding agent can scaffold a three.js project in an afternoon. Spend the saved time on the evaluation. Three 2026 additions: a **visual personal assistant** that sees what you see (brief C4), **AR/VR as the interface** (C9) and **world models that decide in real time** (C10), the direction that DeepMind's Genie 4 and Meta's V-JEPA 2 point to.

Ground rules for 3D studies. Start with one small environment and reuse assets. Hold out layouts, tasks or requests. Keep scene access identical between methods unless access itself is the variable. Score with a separate checker that reads simulator state, not with the model that acted. Record replays so a reader can see the failure, not just the number.

C.1 Research questions: 3D worlds & digital humans

- Does state-aware retrieval of approved scene facts beat a static briefing for an NPC on held-out routes and dialogues, and at what latency?
- Can a camera assistant answer "where did I leave my keys?" better from a timestamped memory than from frames sampled into a long context, and when should it ask which object you mean instead of guessing?
- Does a typed action schema with a geometry checker reduce invalid placements and unintended edits compared with free-form scene edits?
- When the evidence is out of view, how many actions does question-guided exploration save over random or frontier exploration, and which failure remains: not finding, misreading, or stopping too early?
- Does conditioning on audio plus a short style instruction improve beat alignment and blinded style preference over beat-matched retrieval?
- Can a frozen manipulation policy be made robust to perturbations with a small adaptation or a recovery rule, without hurting the unperturbed case?
- How should a speech assistant handle interruptions (barge-in) and memory so that task success rises without end-to-end latency exploding?
- In AR/VR, does giving the assistant the scene graph plus the user's gaze and hand targets reduce wrong placements and clarification questions compared with language alone?
- For a real-time decision task, where does a small latent world model with model-predictive control beat an LLM planner, and does combining them (LLM sub-goals, latent execution) keep the best of both?

C.2 Featured briefs: 3D worlds & digital humans

C1. A GTA-style CUHK-Shenzhen campus with LLM characters Build a playable slice of the campus with a few recognizable landmarks, two NPCs, and three tasks such as finding a room, asking about an event, and following directions. The map can be extruded from OpenStreetMap building

footprints or modelled by hand; the point is the characters, not the architecture. NPCs need a memory of what they saw and said, an action interface (walk to, point at, give directions), and a source of campus facts. For a sense of what one person with coding agents can now build, watch Taipei Rush (台北狂飙, Oct. 2026): a browser GTA-style clone of Taipei built with about 400 parallel agents and more than US$10,000 of tokens, later extended to Hong Kong, Tokyo and other cities. Your campus slice is far smaller; spend the time saved on the characters and the evaluation, and use your own name and assets (Rockstar has had AI clips that reuse GTA material taken down).

Compare. a scripted NPC · an LLM with a static campus briefing · the same LLM with state-aware retrieval of approved campus facts. Both LLM variants get the same landmark queries and game commands.
Measure. factual accuracy, valid actions, quest completion and response latency on held-out routes and dialogue scenarios.
Deliver. a playable browser build, replays of each condition, the held-out scenario set, and the retrieval index.
Week 1: map, landmarks, scripted baseline; Week 2: dialogue + action interface, retrieval; Week 3: held-out tests, latency logs; Week 4: playable build, replays, write-up. Start from: Generative Agents [8] (memory, reflection, planning) · Smallville code · AI Town starter kit · ai-npc-world (three.js village with LLM NPCs) · OpenStreetMap + ExtrudeGeometry for the campus footprint · AgentBench for action-format ideas · video examples: Taipei Rush gameplay (play it in a browser, how it was built) and a GTA6-style China map made with AI (Jack-Cui, Forbidden City; open-source code).

C2. A three.js world with AI-generated assets that adapts to its user Build one browser scene and generate a small library of assets for it with an image-to-3D model, then import the meshes as glTF. The research question is how the scene should adapt to what a user asks for: free-text preferences interpreted by the model, or structured preferences the model fills in.

Compare. a fixed layout · adaptation from free-text preferences · adaptation from structured preferences, all over the same asset library and with the same preview and undo controls.
Measure. fraction of requests satisfied, unintended scene changes, user corrections, asset load failures, frame time.
Deliver. the playable scene, the asset library with generation prompts, and evaluation traces for a fixed request set.
Week 1: asset + preference schema, 10–20 assets; Week 2: two adaptation variants; Week 3: same requests through all variants; Week 4: playable scene, traces, figure. Start from: inspiration post · TRELLIS.2 (MIT, GLB with PBR) · Hunyuan3D 2.1 · three.js glTF loading · HY-World 2.0 for whole scenes (meshes / 3DGS).

C3. A digital human that dances to audio Retarget a small library of motion clips to a single rigged avatar and make it move to any input track. The language angle: a short style instruction ("sharper on the beat", "lazy hip-hop") should change which clips are chosen and how transitions are made.

Compare. fixed playback · beat-matched retrieval · retrieval conditioned on both audio and a style instruction, on held-out tracks.
Measure. beat alignment between movement accents and audio beats, transition jerk or foot sliding, blinded style preference.
Deliver. a browser or recorded demo, 8–12 scored clips per condition, and the scoring scripts.
Week 1: rig, clip library, fixed playback; Week 2: audio + language conditioning; Week 3: score 8–12 clips; Week 4: demo with matched comparisons. Start from: inspiration post · AIST++ and AI Choreographer · EDGE (diffusion dance generation, the standard open baseline) · AtomicDance (ECCV 2026, built on EDGE) · three-vrm and Mixamo for rigs and clips · librosa beat tracking.

C4. A visual personal assistant: it sees what you see, remembers it, and asks which one you mean Smart glasses and phone cameras (Meta's Ray-Ban glasses, Gemini Live with camera sharing, DeepMind's Project Astra) turn the assistant into something that watches your day with you. Build a small one and study a single design choice. Input: a first-person video stream (your own recordings with consent, or clips from EgoLife or Ego4D). State: a memory of timestamped observations. Three kinds of request: about *now* ("what does this label say?"), about *the past* ("where did

I leave my keys?", "how many coffees today?"), and *ambiguous* ones ("is that one cheaper?") where the right move may be to ask which object the user means.

Compare. for memory questions: the current frame only · uniformly sampled frames in a long-context model · retrieval over a timestamped memory of captions and objects. For ambiguous requests: best guess · always ask · ask only when more than one candidate remains plausible after grounding. Same model throughout.
Measure. accuracy on 100–200 held-out questions whose gold answers come from annotated timestamps, latency per answer, memory size and tokens per hour of video; for ambiguous requests, correct selections and questions asked.
Deliver. a phone or browser demo, the question set with timestamps, replays of successes and failures, and a privacy note (who was recorded, what is stored, faces blurred in anything you release).
Week 1: video source, 150 questions, frame-only baseline; Week 2: long-context and memory variants; Week 3: ambiguous set, ask-when-ambiguous trigger; Week 4: demo, replays, write-up. Start from: EgoLife [9] (CVPR 2025; a week of egocentric life video and the EgoLifeQA questions) · Ego4D (episodic-memory queries) · VideoLLM-online [51] (streaming video assistant) · Project Astra · Qwen3-VL · LongLLaVA (our lab, many frames in one pass) · pairs with C16 (when should an assistant ask?) and D1 (long-context failures).

C5. Multi-turn language editing for 3D scenes "Move the chair beside the desk without blocking the door." Compare direct generation of scene edits with a typed action schema whose outputs pass through a geometry checker (collisions, reachability, relation satisfaction). Include follow-up references ("no, the other one") and corrections.

Compare. free-form edit generation · typed schema + geometry checker, same model and same scenes.
Measure. relation satisfaction, unintended edits, invalid placements, turns to success.
Deliver. an editor that reuses existing assets and replays its edit history.
Start from: EditRoom · three.js TransformControls.

C6. Look before you answer: active embodied question answering in a simulated home "Is the stove on?" or "how many mugs are in the sink?" cannot be answered from where the agent starts: it has to decide where to go, what to look at, and when it has seen enough. The task is fixed and small. Use the active split of OpenEQA (HM3D homes in Habitat), or write 60 questions in AI2-THOR whose answers you read from simulator state. The agent is a vision-language model with a short action list (move forward, turn left or right, look up or down, open) and a budget of 20 actions; it ends an episode by answering. The research question is how much a question-guided search strategy saves, and which failure remains.

Compare. answering from the start view (lower bound) · random or frontier exploration, then answer · question-guided exploration, where the model names where the evidence should be and stops when confident · the same plus a semantic map of what it has seen. Same model, same 20-action budget.
Measure. accuracy against simulator state (and OpenEQA's LLM-Match score, checked by hand on a sample), actions used, an efficiency-weighted score, and a split of failures: never saw the evidence, saw it but misread it, or stopped too early.
Deliver. the question set with ground truth, the agent, replays for each condition (top-down path plus first-person frames), and the failure table.
Week 1: simulator, 60 questions, start-view baseline; Week 2: random and frontier exploration, the checker; Week 3: question-guided agent and map, matched budget; Week 4: replays, failure analysis, write-up. Start from: OpenEQA [10] (CVPR 2024; episodic-memory and active splits, LLM-Match metric; code) · Explore until Confident [52] (RSS 2024; a stopping rule for active EQA) · AI2-THOR · Habitat 3.0 · Qwen3-VL as the vision model.

C7. Make a frozen manipulation policy robust (OpenLET / LIBERO) Take an existing vision-language-action policy on a single pick-and-place task in simulation. Do not train a new policy; instead test one small adaptation (a few LoRA steps on perturbed episodes) or one recovery rule (re-grasp on detected failure) against the frozen policy.

Compare. frozen policy · frozen + recovery rule · small adaptation, on seen and perturbed object placements.
Measure. success by placement and perturbation, extra steps used, regressions on the unperturbed

case.
Deliver. evaluation scripts, per-episode logs, and video replays of failures.
Start from: OpenLET Embodied Intelligence Challenge (simulation round to Dec. 31) · LeRobot and SmolVLA-LIBERO · LIBERO · ManiSkill · LLaVA-VLA (a plain VLA baseline meant for students).

C8. A talking digital human: speech-to-speech with barge-in and memory Build a voice assistant for a narrow task (campus directions, a vocabulary drill) with streaming ASR, an LLM, and streaming TTS. The variable is the interaction design: how interruptions are handled and what the assistant remembers across turns.

Compare. turn-taking without barge-in · barge-in with immediate stop · barge-in plus a short memory of the interrupted intent.
Measure. task success, end-to-end latency breakdown (ASR / LLM / TTS), interruptions handled correctly.
Deliver. demo scripts, latency breakdowns, a scripted instruction set with expected outcomes.
Start from: CosyVoice 3 (Apache-2.0, streaming) · Whisper · Silero VAD for barge-in detection.

C9. An LLM assistant inside a WebXR scene: AR/VR editing by voice Lecture 8 asked for an "open world experience using AI + VR". The research version: a user wearing a headset (or using a phone's AR view, or the browser's WebXR emulator) says "put a lamp on the desk, facing the window" and the assistant edits the scene. Microsoft's LLMR (CHI 2024) showed that scene understanding, task planning and self-debugging cut GPT-4's error rate four-fold in Unity; DecartXR (open source, Quest) restyles the room in real time from speech. Your variable is how much of the scene the model gets to see: nothing, a text scene graph, or the scene graph plus the user's gaze and hand targets.

Compare. prompt-only · scene graph in context · scene graph + gaze/hand target, on the same spoken commands and rooms; same model and action schema.
Measure. command success judged by a geometry checker (placement, relation, no collisions), unintended edits, clarification questions, end-to-end latency from end of speech to visible change.
Deliver. a WebXR page that runs on Quest or in the emulator, a replayable command set, and recordings of each condition.
Week 1: scene, action schema, geometry checker; Week 2: three context conditions, voice input; Week 3: held-out commands, latency logs; Week 4: headset recordings, write-up. Start from: LLMR [53] (CHI 2024) and LLMER [54] (JSON instead of code) · DecartXR (open-source real-time restyling on Quest) · WebXR with three.js VR or A-Frame · Immersive Web Emulator if you have no headset · Meta XR Operator (an agent that launches and tests Quest scenes, Aug. 2026) · reuse the checker from C5.

C10. A small JEPA-style world model that decides fast: latent planning vs. an LLM planner Language models plan slowly; a game NPC, a drone or a robot gripper needs a decision every few milliseconds. Meta's V-JEPA 2 (June 2025) learns a video world model by predicting in latent space and plans by rolling out candidate actions with a 300M-parameter action-conditioned head; DreamerV3 and TD-MPC2 are the model-based RL references. The project: in one small environment (PushT, a MetaWorld task, or a 2D game you write), train or fine-tune a compact predictive model and compare a latent planner against an LLM that is asked to choose the next action from a text description.

Compare. a frozen LLM planner (text observations, one call per step) · a small learned world model with model-predictive control · the two combined (LLM sets sub-goals, latent planner executes), at matched wall-clock per episode.
Measure. task success on held-out starts, decisions per second, latency distribution, and how success degrades when the decision budget is cut.
Deliver. the environment, training scripts, replay videos, and a plot of success against decision latency.
Week 1: environment, scripted baseline, LLM planner; Week 2: train the predictive model, CEM/MPC planner; Week 3: matched-latency comparison, ablate horizon; Week 4: combined variant, plots, write-up. Start from: V-JEPA 2 (paper 2506.09985 [55]) · jepa-wms (what drives success in planning with JEPA world models; PushT, MetaWorld, DROID checkpoints) · jepa.cpp (V-JEPA

2-AC on a CPU) · DreamerV3 · TD-MPC2 · DreamDojo (NVIDIA, open-source robot world model at 10 FPS, Feb. 2026) · the CS25 world-modeling talk below.

C.3 Watch, read, play: 3D worlds & digital humans

- Video: **Generative Agents: Interactive Simulacra of Human Behavior** (Park et al., Stanford). The memory stream, reflection and planning design that C1 builds on. Code
- Video: **Taipei Rush (台北狂飙)**: a browser GTA-style Taipei, built by one developer with coding agents in about a month (Oct. 2026). Shows how cheap the map has become for C1; the characters and the evaluation are still your job. Play
- Video: **Genie 3** (Google DeepMind): a real-time interactive world model. Closed, but it sets the bar for what "a world from a sentence" means. Open alternatives: HY-World 2.0, awesome list.
- Video: **Stanford CS25 V6: From Representation Learning to World Modeling**. Background on JEPA-style world models for anyone thinking about Track C beyond the demo.
- **Inspiration posts (X)** three.js scene with AI-generated assets · digital human dancing to sound Design prompts only; the posts were not independently verified.
- **World models in 2026 (X)** Genie 4 (DeepMind, Oct. 27, 2026: real-time steering by text, native audio; research preview) · Runway GWM Worlds 2 (720p, 24 fps, arbitrary actions) · PixVerse R2 (publicly playable) · DreamDojo (open source, robots) · Overworld (local-first, 60 fps) Closed models set the bar; the open ones (DreamDojo, HY-World, V-JEPA 2) are what you can build on. Not independently verified.
- **AR/VR and three.js (X)** DecartXR: speak to restyle your room on Quest · Meta XR Operator lets agents test VR apps · ThreeUI: 160+ open three.js components · three.js Blocks: a 3D site or game from one prompt · three.js r183 tutorial for AI students · three.js at 11M downloads a week: the AI effect Design prompts only.
- **Open 3D and world generation** TRELLIS.2 (MIT, image→GLB) · Hunyuan3D 2.1 · HunyuanWorld 1.0 · HY-World 2.0 (editable meshes / 3DGS) · World Labs World API (hosted) Generated assets garble text, logos and faces; keep landmarks recognizable with hand-made signage.
- **Simulators and embodied benchmarks** AI2-THOR · Habitat 3.0 · ManiSkill · LIBERO · Genesis · MuJoCo · LeRobot The course's Lecture 10 (Multimodal LLMs) covers the vision-language side.

C.4 More research questions: 3D worlds & digital humans

- **Viewpoint-dependent references.** "The cup on my left" changes meaning with the camera. Measure how often a VLM resolves such references correctly from a fixed vs. a user-aligned view.
- **Answer with geometric evidence.** Make a 3D QA system cite the object and the measurement it used; score citation correctness separately from answer correctness.
- **Adaptive screenshot cropping** for GUI or 3D agents: does a model-chosen crop beat a fixed grid at the same token budget?
- **Per-character event memory** for NPCs: compare a flat log, a summarised memory, and a retrieval memory on dialogue consistency over long play sessions.
- **Video captioning or editing with quality controls**, from the multimodal research area: fix the clips and the rubric, vary the control mechanism.

Pitfalls we see every year. A beautiful scene with no held-out test. A single long play session reported as "it works". The acting model also grades the answers. Latency reported without the hardware. Comparing an LLM NPC to a scripted one on tasks the script was never written for. Keep the environment small, the test set fixed, and the checker independent.

Part 2 · Generative AI for education, special audiences & human–AI interaction (briefs C11–C20)

Generation is cheap; **faithful, useful generation is not**. This track takes a concrete piece of teaching material (a textbook passage, one math concept, one molecule, one physics law) and builds a workflow that turns it into something a learner can watch, rotate, or experiment with: an English skit, an explanation video whose narration follows the diagram, an interactive 3D biology or physics scene. The research question is about *source fidelity* (does the output stay true to the material?), *interaction* (does the system know when to ask?) and *transfer* (can the learner apply the idea to a new case?). Human studies are welcome but optional; a scripted evaluation with a transfer quiz is enough for a strong report. Lecture 8 adds audiences beyond the classroom: young children, special groups, specific professions and majors, served by a content pipeline that is **reusable by design and curated by**

hand (brief C19). New this term: a whole **lecture series** generated from one source, with narration and a recurring host (C20).

How to evaluate education projects without a classroom. Pair every generated artifact with (1) a fidelity check against the source (claims and target words mapped to passages), (2) a small blinded rubric for quality, and (3) a **transfer task** on new material (a new sentence, a new graph, a different protein chain). Measure learning as transfer, not as "engagement". If you do recruit classmates, keep it small, anonymous and free of sensitive data, and report the protocol.

C.5 Research questions: education & interaction

- Does a plan that first maps each target word and factual claim to the source produce more faithful and more usable dialogues than a direct script prompt?
- Does a structured storyboard that ties each narrated step to an equation and a visual event reduce narration–diagram mismatches in a math video?
- Does giving the model the learner's selected 3D component and structure annotations improve the correctness and evidence of its answers?
- Do typed simulator commands with trace-grounded explanations beat prompt-only interpretation for "change only the launch angle" requests?
- When should a tutoring assistant ask a clarifying question? Does an ask-when-triggered policy beat always-ask and never-ask on task success and questions asked?
- Do memory and reflection in an educational assistant change satisfaction or learning gain, or only the feeling of continuity?
- What moderation design keeps an LLM-simulated class forum civil without flattening disagreement?
- In a content pipeline for a specific audience, what does ten minutes of human curation per item add that automated checks do not, and which edits recur?
- Does a recurring host and a story that runs across lessons keep learners watching a generated lecture series to the end, without adding factual errors?

C.6 Featured briefs: education & interaction

C11. Textbook passages as short English skits Adapt one textbook passage into a 60–90 second dialogue that teaches four to six target words through context, then voice it with an open TTS model and, optionally, animate it. The Gaoyou duck-egg and WordsFunny posts are format references: a tiny story, a few characters, the words appearing naturally.

Compare. a direct script prompt · a plan that maps each word and claim to the source before writing; same model, target words and running time.
Measure. factual fidelity to the passage, natural use of the words (blinded rubric), and whether learners can use the words in new sentences (transfer quiz).
Deliver. two versions of each skit, transcripts with source mappings, audio or video, and the quiz.
Week 1: passage, word list, fidelity rubric; Week 2: direct vs. planned pipeline, TTS; Week 3: blind rating, transfer quiz; Week 4: videos, write-up. Format references: Gaoyou duck egg · WordsFunny. Tools: CosyVoice 3 (Apache-2.0, cross-lingual cloning) · IndexTTS 2 (emotion control) · Wan 2.2 or LTX-2 for clips · Remotion to assemble subtitles and scenes programmatically.

C12. A math explanation video whose narration follows the diagram Teach one concept, for example why a secant slope approaches the tangent slope. A single curve, two movable points and a line are enough. The variable is the storyboard: unconstrained, or structured so that each narrated step links to an equation and a visual event.

Compare. unconstrained storyboard · structured storyboard, rendered with the same graphics and comparable running time.
Measure. mathematical errors, narration–diagram mismatches (timestamped), and a transfer question on a new graph.
Deliver. two short videos, an editable storyboard, and the scoring rubric.
Start from: Manim Community · Motion Canvas · CosyVoice 3 for narration.

C13. An interactive 3D biology lesson grounded in a molecular structure Teach students to distinguish protein chains from bound ligands and to locate heme in hemoglobin (PDB 4HHB). Learners rotate the structure and ask "which heme is associated with this chain?". The model sees the lesson text in both conditions; only one condition also receives the viewer's selected component and structure annotations.

Compare. lesson text only · lesson text + selected component and annotations, same model and queries.
Measure. correct object identified, relevant structural evidence cited, and whether learners can locate the feature on a different chain.
Deliver. one interactive molecule, annotated queries, a transfer task.
Start from: PDB 4HHB · PDB-101 hemoglobin · RCSB Mol viewer (embeddable).*

C14. A language-controlled 3D physics experiment Teach how launch speed and angle affect projectile range. One scene: a sphere, a ground plane, a trajectory trace. Learners request controlled changes, predict the outcome, run, and compare. Compare a prompt-only interpretation of requests with a model that issues typed commands and explains results from the simulator trace.

Compare. prompt-only · typed commands + trace-grounded explanation, on paraphrased requests.
Measure. valid commands, experiments that change only the intended variable, explanation accuracy on unseen settings, a transfer question.
Deliver. one scene, runnable experiment scripts, and a paraphrase evaluation set.
Start from: Web VPython (GlowScript) · or three.js with a hand-written integrator. Starter assets for other lessons: a pendulum (pivot, rod, bob), or one model from NASA 3D resources with one measurable interaction. Keep asset sources and lesson assumptions in the README.

C15. An AIGC-for-education workflow for one textbook unit, with reusable 3D assets Pick one unit of a real textbook (one chapter of a middle-school science book, one lesson of an English coursebook) and build the pipeline from source to artifacts: a glossary, a short skit, a diagram animation, and one interactive 3D object generated once and reused across lessons. The research question is about the workflow: where does a source-mapping step pay for itself, and where does it not?

Compare. pipeline with and without the source-mapping step, per artifact type.
Measure. fidelity errors per artifact, generation cost, and a short transfer quiz per lesson.
Deliver. the workflow, a lesson pack, and a reusable asset library with provenance.
Combine C11–C14; generate assets with TRELLIS.2; present with Streamlit or Gradio.

C16. When should an assistant ask for clarification? (adapted from Stanford CS329X) Pick one setting: object selection in a visual assistant (see C4), GUI actions, or document editing. Compare immediate execution, asking on every ambiguous case, and asking only when a defined trigger fires. Freeze the model, the task instances, and the information available before the decision.

Compare. execute · always ask · ask on trigger.
Measure. task success, questions asked, corrections; optionally a small human study on perceived helpfulness.
Deliver. a replayable task set and the comparison.
Start from: CS329X: Human-Centered LLMs (Diyi Yang) · Reflections and new directions for human-centered LLMs [56] (the CS329X class paper) · Collaborative Gym.

C17. Memory and reflection in an educational assistant Build a tutoring assistant for one topic and vary only its memory: none, a running summary, or retrieval over past sessions plus a reflection step that updates a learner model. Script learners with known misconceptions so the evaluation is repeatable, then optionally run a small study.

Compare. no memory · summary memory · retrieval + reflection.
Measure. misconceptions addressed, repeated explanations avoided, satisfaction (SUS/UMUX) and a learning-gain quiz if humans take part.
Deliver. the prototype, the scripted learner personas, and interaction logs.
Related: Generative Agents [8] (reflection) · the Kaggle "root misconceptions in K–12 math solutions" task listed on ML Contests for data ideas.

C18. An AI town or class forum: simulating group dynamics and moderation Simulate a class forum with LLM personas and test moderation designs: none, rule-based, or an LLM moderator with a stated policy. The simulation is the test bed; the contribution is a replayable protocol and clear metrics for civility and diversity of opinion.

Compare. moderation designs on the same seeded discussions.
Measure. toxicity, participation balance, opinion diversity, moderator interventions, cost.
Deliver. a replayable simulation and the policy document.
Start from: AI Town · Generative agent simulations of 1,000 people [57].

C19. A reusable AIGC content pipeline for K-12, special groups, or a profession, with a human in the loop Most AIGC demos make one nice artifact. The useful research object is a **pipeline** that makes a hundred, plus the small manual step that turns the good ones into great ones. Choose one audience and one format: a picture book that explains AI to five-year-olds (a Lecture 8 "special topic"), audio-described science stories for blind pupils, simplified-Chinese safety cards for migrant workers, nursing-handover drills for nursing students, Cantonese-opera synopses for primary schools, or case vignettes for a specific major. Build the pipeline as a template (brief → outline → text → image/audio → checks) so a new topic takes minutes, then ask what the human curation step adds.

Compare. fully automatic pipeline · pipeline + automated checks (reading level, source fidelity, safety) · pipeline + checks + 10 minutes of human editing per item, on the same 30 briefs.
Measure. a blinded rubric scored by two people from the audience's world (a teacher, a parent, a nurse); reading-level and fidelity checks; cost and time per item; which edits humans make most (a taxonomy).
Deliver. the templated pipeline, 30 items in each condition, the rubric, the edit taxonomy, and a released content pack with provenance.
Week 1: audience, format, 30 briefs, rubric; Week 2: pipeline + automated checks; Week 3: human-edit condition, blinded rating; Week 4: content pack, write-up. Start from: happy-llm and Happy-Medical-AI (teaching content as open repos) · Bastani et al., PNAS 2025 [23] (unguarded AI help can harm learning: design the checks accordingly) · image models: Qwen-Image, FLUX; TTS: CosyVoice 3 · accessibility references in Track E (E5) · posts on X: Gemini Live "Guided Vision" for blind users, Visionauta, a free app by a blind engineer.

C20. An AI lecture series: generated courseware, TTS narration and a recurring host for one topic Our Lecture 10 page is the worked example: the slides are narrated by a cloned voice and a photo avatar, a laser pointer follows the narration, and a chapter panel and lecture map sit beside the video. Turn that into a pipeline and make a **series**. Choose one topic (probability through card games, the history of Chinese characters, how vaccines work, LLMs for high-school students) and one source (a textbook chapter, a set of notes), and generate four to six connected lessons: outline, slides, script, narration, pointer cues and a short quiz per lesson. A recurring host makes a series feel like one: an original mascot, a figure from public-domain literature drawn in your own style (孙悟空, 诸葛亮), or a character whose rights you hold. Do not use commercial characters, voices or faces without permission.

Compare. slides with TTS narration · plus synchronised pointer and highlight cues · plus the host character and a story that runs across lessons; same source, same lesson length. Optional fourth condition: ten minutes of teacher editing per lesson.
Measure. factual errors against the source, narration–slide mismatches (timestamped), a quiz and one transfer question per lesson for 10–20 learners, completion across the series (who reaches lesson 4?), and cost and minutes per finished lesson.
Deliver. the pipeline as a template (a new topic in under an hour), the series as a web page or playlist, scripts with source mappings, the quiz data, and a note on consent and rights for every voice, face and character used.
Week 1: topic, source, series outline, one lesson made by hand as the reference; Week 2: pipeline: slides, script, TTS, pointer cues; Week 3: host character, all lessons, learner quiz; Week 4: series page, completion data, write-up. Start from: Lecture 10 AI-video page (cloned voice, avatar, laser pointer, chapters) · Paper2Video [12] (slides → subtitles → speech → cursor → talking head; code) · PPTAgent [58] · From MOOC to MAIC [11] (Tsinghua's AI-run courses) · TTS: CosyVoice 3, IndexTTS 2 · talking heads: LivePortrait, Hallo · slides as code: Slidev, reveal.js, Manim (see C12).

Clone only your own voice and face or those of people who agreed in writing, and label the videos as AI-generated, as China's 2025 rules on labelling AI-generated content require for public posts.

C.7 Watch, read, play: education & interaction

- Video: **Generative Agent Simulations of 1,000 People**: how interview-grounded personas were built and validated. Relevant to C17 and C18.
- **Stanford CS329X: Human-Centered LLMs** (Diyi Yang, Fall 2026) course site with public slides on human preferences, RLHF/DPO and design thinking · 60-page class paper [56] on design, data sourcing, training, evaluation and deployment Suggested project categories there: apply an LLM to a real problem; new ways to use human feedback; fairness and bias; interactive systems; persona simulation.
- **AI tutors in 2026 (X)** StudentBench: AI and human tutoring yield equivalent GRE gains (n=2,383; paper [59], data on Hugging Face) · Harvard physics crossover study: AI tutor vs. active-learning class · Brilliant's Koji, a real-time STEM tutor · Aristotle: a tutor that refuses to give answers, and why that is a product bet Read with Bastani et al. (PNAS 2025) [23]: AI help without guardrails raised practice scores and lowered exam scores.
- **Format references (X)** Gaoyou duck egg · WordsFunny Design prompts only; not independently verified.
- **Open generation stacks for lessons** TTS: CosyVoice 3, Qwen3-TTS, IndexTTS, Fish Speech (check licenses) · video: Wan 2.2, HunyuanVideo 1.5, LTX-2 · animation: Manim, Remotion, Motion Canvas
- **Interactive science viewers** Mol* · GlowScript / Web VPython · NASA 3D resources · three.js

C.8 More research questions: education & interaction

- **Task-oriented UX for grounded report writing** with inline citations: does showing the retrieved span next to the claim change how often users accept wrong claims? (overlaps with D9)
- **How users interrupt a task**: log interruption points in a long-running assistant and test two resumption designs.
- **Pointing to resolve ambiguity** in a 3D editor: compare language-only with language + pointer on the same ambiguous commands.
- **Persona simulation for pilot studies**: can simulated learners predict which of two explanations real learners prefer? Measure agreement on a small held-out set.

Pitfalls we see every year. Measuring "engagement" or "likes" instead of learning. Reporting a beautiful video with no fidelity check against the source. No transfer task, so there is no evidence of learning. Human studies that are too small to say anything and too large to finish. Keep the artifact small and the comparison sharp.

D Evaluation, datasets & evidence

Measurement and data are two halves of the same craft: a benchmark is a dataset with a scoring rule, and a dataset is only as good as the evaluation that shows what it changes. Part 1 builds evaluations that can show a specific failure of today's strongest models (culture, humour, logic, long context, injection, machine-written text) and scores them with rubrics written by people who know the domain. Part 2 builds the artifacts that training and retrieval depend on: datasets in the format each stage expects, corpora with provenance, evidence chains a reader can audit, and corpora that AI helped construct and a verifier checked. Every brief here runs on an API model or a CPU, and every one ends with a released artifact and a matched comparison.

Part 1 · Evaluation: where frontier models still fail, rubrics & safety (briefs D1–D8)

Lecture 8's third topic is evaluation, and its list is still a good map of where the strongest models stumble: Chinese culture and idioms, regional stereotypes, sense-making, formal logic, humour, vision and speech, long context, emotional intelligence, multi-turn dialogue, ethics. In 2026 the twist is that the classic benchmarks are saturated and contaminated, so the valuable contribution is **a small, fresh test where you can show a specific failure**, scored in a way a reader can trust: executable checks, ground truth, or **rubrics** written by people who know the domain (HealthBench's 262 physicians are the template). The track also covers the safety side of evaluation: prompt injection in agents, detecting machine-written text, verifiers that accept unearned success. Pick one failure and one domain, and build the smallest evaluation that measures it.

Evaluation discipline. Include unanswerable and ambiguous cases, so abstention can be measured. Collect a human baseline from five to ten classmates; a benchmark without one cannot say "easy for humans". Audit a random sample by hand and report agreement with the automatic metric. Never let the generator judge itself; if an LLM judge is unavoidable, use a different model and validate it against the audited sample. Check for contamination by asking each model to continue a truncated item. Keep intermediate logs replayable.

D.1 Research questions: evaluation

- On a fresh suite in one of the Lecture 8 areas (culture, humour, logic, EQ, dialect, multi-turn), where do three frontier models and one open model fail, and does the failure survive paraphrase and translation?
- Does a rubric judge agree with domain experts more than a holistic 1–10 judge, and do automatically generated rubrics close the gap with hand-written ones?
- Does a long-context evaluation that mixes "needle" retrieval with multi-step reasoning expose failures that needle tests alone miss, and do the same failures appear when the context is hundreds of images instead of text?
- Do combined defenses (input sanitization + sandbox + policy model) reduce prompt-injection success in a web agent without blocking legitimate tasks?
- How separable are human-written, LLM-generated and LLM-refined texts, and which features survive a change of generator?
- With a few hundred queries, can you tell which model (or which mixture of models) a relay API is really serving behind the name it advertises?
- Can a system tell whether an AI-written scientific statement faithfully reflects the evidence it cites?
- How often does an LLM verifier accept an unsupported success claim, and does the bias depend on who wrote the claim?

D.2 Featured briefs: evaluation

D1. A long-context evaluation that detects a specific failure Design a small long-context suite where retrieval and reasoning are separable (find the facts; then combine them), run it across context lengths and models, and show which failure appears first. Text is the default. The multimodal version puts the facts in images: one relevant image among 100–1,000, or two images that must be compared. Our lab's LongLLaVA, a hybrid Mamba–Transformer model that reads about 1,000 images on one

80 GB GPU, is a baseline to beat or to analyse, and MileBench supplies multi-image tasks to start from.

Compare. models or context lengths on the same suite; needle-only vs. needle + reasoning items; for the multimodal version, the same items as text vs. as images, and a hybrid model (LongLLaVA) vs. a Transformer model at matched memory.
Measure. accuracy by position and length, failure type, cost; for images, also throughput and peak GPU memory as the image count grows.
Deliver. the suite with seeds and replays.
Start from: LongBench · lm-evaluation-harness · LongLLaVA [14] (our lab; code and weights) · MileBench [15] (our lab; long-context multi-image tasks).

D2. Combined defenses against prompt injection in a web agent Take an agent benchmark with injected instructions in tool outputs and test defenses alone and combined: input sanitization, a sandboxed tool layer, a policy model that screens actions.

Compare. no defense · each defense · combined, on the same tasks and attacks.
Measure. attack success rate, benign task success (utility), false positives, cost.
Deliver. threat model, attack set, defense configs, results.
Start from: AgentDojo · Weak-to-Strong Jailbreaking [60].

D3. Who wrote this? Detecting and attributing LLM-generated text Two live or archived tasks fit: Kaggle's *Name That LLM* (identify which of 18 models wrote a passage; offline notebooks, nine-hour limit; entry Jan. 6, final Jan. 13, 2027) and NLPCC 2026 Shared Task 6 (three-way: human-written, LLM-generated, LLM-refined; Chinese news and academic text). Compare character n-grams with a small encoder on held-out prompts, and test transfer across generator families.

Bonus question: which model is really behind the API? Attribution has a practical twin. Relay services (中转站, "shadow APIs") resell access to frontier models, and a 2026 audit found that 11 of 24 such endpoints failed fingerprint verification; one sold as GPT-5 matched GLM-4-9B. Given an endpoint that claims to serve model X, can you tell whether it does, or estimate the mixture behind it (say 70% X and 30% a cheaper model)? Collect reference outputs from the official APIs, then compare a two-sample test on output distributions (model equality testing), fingerprint prompts plus a per-response classifier with a mixture estimate, and metadata cues (token counts, latency, refusal style, knowledge cutoff). Validate everything first on a relay you build yourself that mixes known models in known proportions. Test only endpoints you pay for, within their terms, and report a mismatch as evidence, not as proof of intent.

Compare. n-gram features · a fine-tuned small encoder · an ensemble, with a held-out generator family.
Measure. balanced accuracy or macro-F1, confusion across families, robustness to paraphrase.
Deliver. the notebook, features, and a confusion analysis.
Start from: Name That LLM · NLPCC 2026 Task 6 · Idiosyncrasies in large language models [61] (which model wrote it, from text alone) · for the bonus: Real Money, Fake Models [17] (shadow-API audit) · Are you getting what you pay for? [62] (model substitution) · Model equality testing [16] · LLMmap [63] (fingerprinting) · rank-based uniformity test [64] · API-TAIL [65] (partial substitution).

D4. Reliability of AI-assisted scientific reporting NLPCC 2026 Shared Task 10 asks: given scientific evidence and an AI-generated statement, does the statement faithfully reflect the evidence it summarizes or cites? Track 1 is claim-level faithfulness to experimental results; track 2 is citation-level faithfulness. The data and metrics are public; run it locally with your own comparison.

Compare. a zero-shot judge · a judge with retrieved evidence spans · a fine-tuned classifier.
Measure. sentence macro-F1 and paragraph exact match (track 1), label macro-F1 and Joint@3 (track 2).
Deliver. the system and an error analysis by claim type.
Start from: NLPCC 2026 Task 10.

D5. Auditing verifier bias and unsupported success claims Many agent results depend on an LLM verifier. Collect trajectories from one benchmark, have humans label success, and measure how

often the verifier accepts unsupported claims ("I have fixed the bug") and whether its bias depends on the generator model or on the phrasing of the final message.

Compare. verifiers (model, prompt, with/without execution evidence) against human labels.
Measure. false-positive and false-negative rates, agreement, bias by generator.
Deliver. the labelled set and a verifier recommendation.
Related: Inspect AI · Predictive AI Evaluation Competition (NeurIPS 2026).

D6. Steerability: does the model do what the instruction asked, and only that? The NeurIPS 2026 Steerability Challenge measures whether a model can be moved to a requested target in a multi-dimensional output space without side effects. Build a small steerability suite for one domain (tone, length, reading level, citation style) and compare prompting strategies or a lightweight fine-tune.

Compare. direct instruction · decomposed instructions · a steering fine-tune.
Measure. target attainment, side effects on untouched dimensions, consistency across seeds.
Deliver. the suite and results.
Start from: The Steerability Challenge · AlignBench (Chinese alignment evaluation).

D7. Where do frontier models still fail? Build a small benchmark they cannot pass yet Take one area from the Lecture 8 list (Chinese idioms and 歇后语, regional stereotypes, humour, formal logic, sense-making, EQ, speech or vision, long multi-turn dialogue, ethics) or one the 2026 leaderboards point to: tasks that are easy for people and hard for models (SimpleBench, ARC-AGI-3 interactive environments), predicting how specific cohorts behave (Kaggle's Human Behavior Bench), dialect speakers (Hofmann et al., Nature 2024), cross-cultural everyday knowledge (Naous et al., ACL 2024), long-horizon business tasks (Vending-Bench, TheAgentCompany). Write 100–300 fresh items with gold answers or rubrics, collect a human baseline, and run four models. The result is the failure taxonomy, not the leaderboard.

Compare. three frontier API models and one strong open model, with and without chain-of-thought, on the same items; the human baseline from 5–10 classmates; a paraphrased or translated copy of the suite to test robustness.
Measure. accuracy and calibration, agreement with humans, failure categories (knowledge, reasoning, instruction, refusal, format), contamination signal (can the model continue a truncated item?), cost per item.
Deliver. the dataset with a license and data card, the evaluation harness, the human-baseline protocol, and a one-page error analysis per model.
Week 1: area, item template, 50 pilot items, human pilot; Week 2: full suite, gold answers or rubrics; Week 3: run models, human baseline, robustness copy; Week 4: taxonomy, data card, write-up. Start from: the Lecture 8 slide lists Fresh bench, a speech-to-speech bench, AV-Odyssey, a user-centric benchmark, MedQA-pro, Patient-O1, Omni-MATH, MileBench, MAMO, ACVA and MLLM-Bench as examples (search Google Scholar) · Humanity's Last Exam [66] · ARC-AGI-3 · dialect prejudice (Nature 2024) [18] · Having Beer after Prayer? [67] (cultural bias, ACL 2024) · AlignBench (Chinese) · posts on X: Mollick on human baselines, Human Behavior Bench, frontier models failing at nonlinear PDE solvers, expert re-grading: often the benchmark is broken, not the model, contamination and "benchmaxxing".

D8. Rubric-based evaluation: write rubrics, measure judge agreement, then use them as rewards A rubric decomposes "is this answer good?" into criteria with points: HealthBench pairs 5,000 health conversations with 48,562 physician-written criteria and finds model–physician agreement close to physician–physician agreement. Rubrics then became rewards: *Rubrics as Rewards* (Scale AI, 2025) trains with checklist feedback beyond verifiable domains, *Reinforcement Learning with Rubric Anchors* scales rubric construction, *OpenRubrics* generates rubrics by contrasting preferred and rejected answers, and *Checklists are better than reward models* makes the case against scalar rewards. The 2026 follow-ups (ImpossibleRubrics, reward hacking in rubric RL) show what goes wrong. Pick one open-ended domain you can get two experts for (a medical sub-specialty, legal drafting, Chinese essay grading, code review comments, teaching feedback) and run the full chain.

Compare. a holistic 1–10 judge · a judge with hand-written rubrics · a judge with automatically generated rubrics (contrastive generation), all on the same 100 responses from four models, against two expert raters; optionally use the best rubric as a reward for a short GRPO run (Tutorial 2) and check the policy on a held-out set.

Measure. agreement with experts (Cohen's κ per criterion, Spearman overall), expert–expert agreement as the ceiling, discrimination between models, judge cost; for the RL step, held-out rubric score, a generic quality check, and any criterion the policy learned to game.
Deliver. the rubric set with authorship, the 100-response gold set, the agreement tables, and a list of criteria that judges apply inconsistently.
Week 1: domain, 100 responses, expert recruitment; Week 2: hand-written and generated rubrics; judges; Week 3: expert ratings, agreement analysis; Week 4: optional GRPO run; write-up. Rubric references: HealthBench [13] (OpenAI, 2025; post) · Rubrics as Rewards [68] · RL with Rubric Anchors [69] · OpenRubrics [70] (data) · Checklists are better than reward models [71] · RubricHub [72] · ImpossibleRubrics [73] · Reward hacking in rubric-based RL [74] · PaperBench (rubric trees) · Prometheus [75] (ICLR 2024) and Prometheus 2 [76] · MT-Bench / LLM-as-a-judge [77] (NeurIPS 2023) · Fine-grained human feedback [78] (NeurIPS 2023) · G-Eval [79] · FLASK [80] (ICLR 2024) · Rubric-based rewards for RL (overview).

D.3 Watch, read, play: evaluation

- **Evaluation tooling** lm-evaluation-harness · Inspect AI · simple-evals (HealthBench graders) · LongBench · AlignBench · AgentDojo · OpenRouter for running many models through one API
- **Where models fail, 2026 (X)** benchmarks are losing their human baselines · Human Behavior Bench: fine on the average American, poor on specific cohorts · physics re-grading finds broken evaluations · what the weekly benchmark scores actually test · METR's task-horizon update Design prompts only; not independently verified.
- **NeurIPS 2026 evaluation competitions** Steerability Challenge · Predictive AI Evaluation · QuantiPhy (quantitative physical reasoning for VLMs) · SimulacraBench (synthetic humanitarian surveys, closed evaluation) Check each site for the submission date; the competition track meets Dec. 11–12.
- **Chinese shared tasks with public data** NLPCC 2026 shared tasks (Task 1 medical instructional video QA, Task 6 LLM-text detection, Task 9 AI Scientist Benchmark, Task 10 scientific reporting reliability) · CCKS 2026 evaluation tasks on Tianchi (cultural-tourism knowledge graph; LLM behaviour control)
- **Rubric reading list** HealthBench [13] · Rubrics as Rewards [68] · OpenRubrics [70] · Checklists > reward models [71] · Prometheus [75] · MT-Bench [77] · ImpossibleRubrics [73] The instructor's rubric survey groups these into evaluation, reward modelling, multi-aspect alignment, LLM-as-a-judge and constitutional methods; ask for it in office hours.
- **Course material** Lecture 8, Topic 3: evaluation · Lecture 9: Medical LLMs · Agentic Hospital (our lab)

D.4 More research questions: evaluation

- **Speech and vision versions** of a text failure: does the same model fail the same item when it is spoken or shown in a photo?
- **EQ and multi-turn**: a scripted user with a hidden emotional state; does the model notice, and does noticing change its answers?
- **Memory compression** in long conversations: what is lost first, facts or commitments?
- **Risk, benefit and disclosure design** in a regulated application: what must the interface say, and does saying it change user behaviour?
- **Robustness under controlled input perturbations** (typos, paraphrase, language mixing) for any system on this page.

Pitfalls we see every year. The same model generates and judges. No human baseline, so "hard" means nothing. No unanswerable questions, so abstention is never tested. Items copied from an existing benchmark, so contamination explains the result. A rubric with twenty criteria and no inter-rater agreement. Audit a sample by hand and say how big it was.

Part 2 · Datasets, corpora & evidence (briefs D9–D14)

Models come and go; a good dataset is cited for years. Lecture 8's fourth topic is dataset building, and each training stage wants a different artifact: documents for pre-training, instruction–response pairs for SFT, chosen/rejected pairs for preference optimization, prompts with verifiers for RL, trajectories for agents, rubrics for judges. This track also covers **evidence**: corpora whose every passage has provenance (source, date, version, span), retrieval pipelines that can show their work, and corpora

that AI helped construct and that a verifier checked. The output is a released dataset or corpus with a data card, a loader, a validation script, and one experiment that shows what it changes.

Data discipline. Record the license of every source and respect it. Strip personal data and say how. Deduplicate against the evaluation sets you will use, and report the overlap you removed. Keep provenance per item (where it came from, which prompt or person produced it, which checks it passed). Write the data card before the experiment, not after. A dataset nobody can load is not a deliverable; a validator that rejects malformed rows is.

D.5 Research questions: datasets & evidence

- At matched size, does a dataset built by your pipeline beat an existing public one for the same stage on a small model, and which step of the pipeline accounts for the difference?
- Does verifying each answer claim against a cited source span reduce unsupported answers at an acceptable cost in coverage?
- Which component of an agentic RAG pipeline (query planning, retriever ensemble, draft-and-critic, citation repair) moves groundedness most, per unit of latency?
- When should a system retrieve at all, and what happens with conflicting or stale sources?
- How much of an AI-synthesised corpus is wrong before verification, how much after, and does training or retrieving over it still help at matched tokens?
- Two deduplication or quality-filtering policies at the same output size: which yields the better downstream model, and what did each throw away?

D.6 Featured briefs: datasets & evidence

D9. Auditable evidence chains for medical or legal QA Choose one dated public medical guideline collection or one legal corpus. Build a replayable pipeline that records, for each claim, the retrieved passage, the document version, the exact supporting span, and the verifier's decision. Compare ordinary retrieval-augmented generation with the same retriever and model plus span verification and abstention.

Compare. RAG · RAG + span verification + abstention, same retriever and model.
Measure. span recall, claim support, citation completeness, error rate among answered questions, coverage; on 50–100 held-out questions including ambiguous and unanswerable ones; manual audit of a sample.
Deliver. a source-inspection demo using public evidence and offline scenarios.
Week 1: corpus, 50–100 questions, baseline RAG; Week 2: span verifier + abstention; Week 3: matched runs, manual audit; Week 4: inspection demo, write-up. Start from: ALCE · LegalBench-RAG [81] · clinical guideline QA benchmark · our lab's HuatuoGPT-II [44] and Lecture 9.

D10. An agentic RAG pipeline with a regression suite Implement the workflow pieces one at a time: query planner (decomposition), retriever ensemble, draft and critic, citation repair; then a regression suite of gold questions over a versioned corpus so that every change is measured. The research result is an ablation across components.

Compare. classical RAG vs. each added component, at matched latency where possible.
Measure. groundedness, citation correctness, regression pass rate, end-to-end latency.
Deliver. the pipeline, the suite, and the ablation table.
Start from: GraphRAG (as one retriever) · DSPy · LangGraph · Tutorial 4 (RAG and agent development, Nov. 13).

D11. Retrieval triggers, conflicting sources, stale documents Construct a corpus with deliberate version conflicts (an old and a new guideline) and questions whose answers changed. Test when the system retrieves, how it resolves conflicts, and whether it reports the version it used.

Compare. always retrieve · retrieve on uncertainty · retrieve with version-aware ranking.
Measure. correct-version rate, conflict detection, unnecessary retrievals, latency.
Deliver. the versioned corpus and the test set.

D12. A reproducible data pipeline with deduplication and quality scoring Build a web or text cleaning pipeline (extraction, language ID, deduplication, quality scoring, PII and toxicity checks) and evaluate the choices downstream on a small training run or an existing model's fine-tune. Alternatives from the research area: construct a small preference dataset and study its effect on instruction following; compare synthetic vs. curated human data for one task; select the hardest SFT examples; audit train–test overlap.

Compare. two deduplication or filtering policies at the same output size.
Measure. coverage and quality scores, downstream gain, PII and toxicity leakage.
Deliver. scripts, data cards (sources, licenses, risks), ablations.
Start from: datatrove · Dolma toolkit · synthetic data for instruction tuning [82].

D13. Build and release a training dataset for one stage, in the format that stage expects Choose the stage first, because the format and the quality checks follow from it. Then choose a domain or audience nobody has covered well (a Chinese dialect, a vocational trade, a university's regulations, a hospital department's FAQs, a game engine's API, your own course's lecture notes) and build 2k–50k items with provenance. The Lecture 8 examples from the lab are Huatuo-26M, MedPajama and VLFeedback; the 2026 open releases to imitate are Tulu 3, HelpSteer, DAPO-Math and the K2 Horizon data drop. Validate the dataset by training: a small model on your data against the same model on an existing dataset of the same size.

- **Pre-training / continued pre-training**. *Row format:* One document per row: `{text, source, url, date, license}`; *Quality checks that matter:* Language ID, exact + near deduplication, quality classifier, PII removal, contamination against eval sets, mixing ratios; *Public examples:* FineWeb-Edu, Dolma, Nemotron-CC
- **SFT**. *Row format:* `{messages: [{role, content}, ...]}` with a chat template; or prompt/response pairs; *Quality checks that matter:* Response correctness, instruction diversity (taxonomy), length balance, decontamination, licence of the teacher model's outputs; *Public examples:* Tulu 3 SFT, Magpie, Huatuo-26M
- **Preference (DPO, reward model)**. *Row format:* `{prompt, chosen, rejected}` or ranked lists with rater IDs and rationale; *Quality checks that matter:* Inter-rater agreement, position and length bias, rubric used, where the pairs came from; *Public examples:* HelpSteer2, UltraFeedback, VLFeedback
- **RL with verifiable rewards**. *Row format:* `{prompt, answer, verifier}` where the verifier is code or an exact-match rule; *Quality checks that matter:* Verifier precision on a hand-checked sample, difficulty spread, no leakage from eval, unambiguous answers; *Public examples:* DAPO-Math-17k, SWE-Gym
- **Agent trajectories**. *Row format:* Prompt + tool schemas + turns of `{action, observation}` + outcome label; *Quality checks that matter:* Executable replay, outcome verified by tests, failure cases kept, environment version pinned; *Public examples:* ToolBench, SWE-smith
- **Rubric / judge data**. *Row format:* `{prompt, response, rubric: [{criterion, points, met}]}`; *Quality checks that matter:* Who wrote the criteria, agreement between judges, criteria that never fire; *Public examples:* HealthBench [13], OpenRubrics

Compare. your dataset vs. an existing dataset of the same stage and size, same base model and training recipe; plus an ablation that removes your main quality check.
Measure. downstream task score, contamination rate, validator rejection rate, cost per item, and a data-card checklist score (sources, licenses, risks, intended use).
Deliver. the dataset on Hugging Face with a card, the generation and validation scripts, and the matched-size comparison.
Week 1: stage, domain, format validator, 500-item pilot; Week 2: scale to the full set; checks; Week 3: matched-size training comparison; Week 4: card, release, write-up. Start from: datatrove · Dolma toolkit · dataset cards · synthetic data for instruction tuning [82] · K2 Horizon's open pre- and post-training data pipelines (Oct. 2026) · Tutorial 2 for the training check.

D14. An evidence-based corpus for RAG, and an AI-built corpus with a verifier Lecture 8 asks you to "replicate OpenEvidence": a medical answer engine whose every sentence points at a guideline or trial. Part one builds the corpus such a system needs, in a domain you can access: one society's guidelines, a university's regulations, a city's public-health notices, a game's wiki. Every

chunk carries source, date, version and section, and a question set with gold evidence spans measures retrieval. Part two asks whether AI can **construct** corpus text: rephrase documents into textbook style (Nemotron-CC, Cosmopedia), synthesise documents from a knowledge graph (EntiGraph), or let an agent build data (Meta's Autodata, the Hugging Face synthetic-data playbook), with a verifier that checks every generated claim against the source corpus.

Compare. raw corpus · raw + AI-rephrased · raw + AI-synthesised without verification · the same with the verifier, at matched tokens; evaluated as a RAG knowledge base (and optionally as continued pre-training data).
Measure. retrieval recall of gold spans, grounded answer accuracy, unsupported-claim rate, synthetic-claim error rate before and after verification, cost per thousand tokens produced.
Deliver. the corpus with provenance, the question set with evidence spans, the verifier, and the four-condition table.
Week 1: sources, chunking with metadata, 100 questions with spans; Week 2: rephrasing and synthesis pipelines; Week 3: verifier; four-condition RAG runs; Week 4: error analysis, release, write-up. Start from: OpenEvidence (the target) · Cosmopedia · Nemotron-CC synthetic rephrasing · EntiGraph: synthetic continued pre-training [83] · posts on X: the Hugging Face synthetic-data playbook (90 experiments, 1T tokens), Meta's Autodata: an agent as data scientist, the lineage of synthetic data · ALCE for citation evaluation · pairs with D9.

D.7 Watch, read, play: datasets & evidence

- **Data tooling** datatrove · Dolma · NeMo Curator · distilabel (synthetic data and feedback pipelines) · GraphRAG · DSPy · LangGraph · Tutorial 4 (RAG and agent development, Nov. 13)
- **Data releases and pipelines, 2026 (X)** K2 Horizon: 13T pre-training tokens and the generation pipelines, open · Hugging Face synthetic-data playbook · Meta Autodata · on the lineage of synthetic data · IrokoBench: human-translated benchmarks for 16 African languages · CulturaX, 167 languages Design prompts only; not independently verified.
- **Lab datasets to study** Huatuo-26M · VLFeedback · BlendNet (CAD scripts) · FreedomIntelligence on Hugging Face Read their data cards and ask what you would add.
- **Course material** Lecture 8, Topic 4: dataset building; Topic 9: retrieval and RAG · Lecture 9: Medical LLMs · HuatuoGPT [43] · Agentic Hospital

D.8 More research questions: datasets & evidence

- **Better retrievers for one domain** (AI for math, a legal code): do synthesised hard negatives improve a small embedding model more than more positives?
- **Hallucination detection and warning** at the sentence level: which signal (retrieval disagreement, self-consistency, entropy) warns earliest at a fixed false-alarm rate?
- **Scene constraints and collisions** as an evaluation corpus for 3D editors (pairs with C5).
- **Selecting the hardest SFT examples**: does training on the 20% the model gets wrong beat training on all of them at matched steps?
- **Auditing train–test overlap** in a popular public dataset and publishing the cleaned split.

Pitfalls we see every year. A dataset with no license and no provenance. Teacher-model outputs released under a licence the teacher forbids. Synthetic text that nobody checked, so the "improvement" is contamination. A citation metric that checks format, not support. Keep the validator, the card and the overlap report in the repository.

E AI for society: position papers, audits & interventions

AI is reshaping work, language, education and public services faster than the evidence can keep up. This track asks you to take a clear position on one question about that future and to back it with a measurement you ran yourself. The questions Lecture 8 raised are the ones below: does AI **widen gender inequality**; does it **reduce the voice of minority languages and dialects**; does it **cut or extend the welfare of people with disabilities**; does it **leave the Global South behind**; does it **concentrate wealth** or spread productivity? The deliverable can be a position or perspective paper in the style of the ICML and NeurIPS position tracks (a claim, the evidence for and against, the alternative views, and what should change), an audit study with a controlled design, or an intervention that measurably helps a group. All three are graded on the same thing: a falsifiable claim and an honest test of it.

What a position paper needs. A one-sentence claim that someone could disagree with. The strongest version of the opposing view, stated fairly. Evidence from the literature, organised by theme rather than listed (the "related work is not a shopping list" rule from the writing tips). One piece of evidence you produced: an audit, a replication, a survey, a cost calculation. A concrete recommendation with who should act. ICML's position track asks reviewers to judge "is the argument well-supported and does it advance the discussion?", not "is the method novel?".

E.1 Reading by question (top-venue starting points)

- **Gender inequality**. *Papers to start from:* Wan et al., EMNLP Findings 2023 (gender bias in LLM-written reference letters) · Blodgett et al., ACL 2020 (what "bias" in NLP means) · UNESCO 2024 (bias against women and girls in LLMs) · Bolukbasi et al., NeurIPS 2016 · Caliskan et al., Science 2017; *What you could measure:* Recommendation letters, hiring summaries, salary advice or career suggestions generated for matched male/female personas in Chinese and English; who gets "warm" and who gets "competent".
- **Minority languages and dialects**. *Papers to start from:* Hofmann et al., Nature 2024 [18] (dialect prejudice in decisions) · Ahia et al., EMNLP 2023 (do all languages cost the same?) · Joshi et al., ACL 2020 (the state and fate of linguistic diversity) · Blasi et al., ACL 2022 (systematic inequalities across languages) · Naous et al., ACL 2024 [67] (cultural bias) · Nekoto et al., Findings 2020 (Masakhane participatory research) · NLLB Team, Nature 2024; *What you could measure:* Token cost, latency and accuracy for 粤语, 闽南语, 藏语, 维吾尔语, 壮语 or your hometown dialect against Mandarin and English; whether a model's advice changes when the same question is asked in dialect.
- **People with disabilities**. *Papers to start from:* Gadiraju et al., FAccT 2023 (disability-centred perspectives on LLMs) · Hutchinson et al., ACL 2020 (social biases as barriers) · Glazko et al., FAccT 2024 (disability bias in GPT-based resume screening) · ASSETS 2024 (how users with disabilities use chatbots) · Be My AI and Gemini Live Guided Vision as the benefit side; *What you could measure:* Resume screening with and without a disability-related award; image descriptions rated by blind users; sign-language or dysarthric-speech recognition accuracy against typical speech.
- **The Global South**. *Papers to start from:* Sambasivan et al., FAccT 2021 (re-imagining algorithmic fairness in India) · Mohamed, Png & Isaac, 2020 (decolonial AI) · Ahia et al., Findings 2021 (the low-resource double bind) · Durmus et al., 2023 [84] (GlobalOpinionQA) · Santurkar et al., ICML 2023 (whose opinions do LMs reflect?) · Kapoor et al., ICML 2024 position (open foundation models); *What you could measure:* Health, agriculture or legal advice for a Kenyan, Bangladeshi or Peruvian user versus a Californian one, scored by people from each place; which countries' sources a model cites.
- **Wealth, work and inequality**. *Papers to start from:* Eloundou et al., Science 2024 (GPTs are GPTs) · Noy & Zhang, Science 2023 (productivity effects) · Brynjolfsson, Li & Raymond, QJE 2025 (generative AI at work: novices gain most) · Brynjolfsson, Chandar & Chen, 2025/26 (early-career employment in exposed jobs) · Humlum & Vestergaard, NBER 2025 (small labour-market effects so far) · Anthropic Economic Index · Bastani et al., PNAS 2025 [23] (AI without guardrails harms learning) · Doshi & Hauser, Science Advances 2024 (creativity up, diversity down); *What you could measure:* Who gains from an AI assistant on a real task: novices or experts, in your own class; whether AI-written outputs converge on the same ideas; task exposure of the jobs your classmates want.

- **Pluralism and whose values**. *Papers to start from:* Sorensen et al., ICML 2024 position (a roadmap to pluralistic alignment) · Bender et al., FAccT 2021 (stochastic parrots) · Weidinger et al., FAccT 2022 (taxonomy of risks) · Kirk et al., Nature MI 2024 (personalised alignment) · Kleinberg & Raghavan, PNAS 2021 (algorithmic monoculture) · Costello et al., Science 2024 (AI dialogues reduce conspiracy beliefs); *What you could measure:* Whether alignment reduced the diversity of opinions a model will voice; whether a model persuades students in your class on a contested campus issue, and how durably.

E.2 Research questions this track asks

- Does the same request, asked by matched personas that differ only in gender, dialect, disability or country, get systematically different answers, and does the difference survive a change of model and language?
- How much more do speakers of a Chinese dialect or a minority language pay in tokens, latency and errors for the same service, and does a tokenizer or prompt intervention close the gap?
- Who gains more from an AI assistant on a real task, novices or experts, and does the gain persist when the assistant is removed?
- Does an accessibility feature built on a multimodal model (image description, speech for dysarthric users, sign-language glossing) help the intended users on tasks they chose?
- Which of a model's answers would change if it were trained or retrieved on sources from the Global South, and can you show it with a small retrieval intervention?
- When a position paper argues from the literature, which claims are supported by replicated evidence and which rest on a single study?

E.3 Featured briefs

E1. A position paper with an audit behind it: "AI will widen (or narrow) X unless Y" Choose one of the six questions above and write a six-page position paper in the ICML position-track format: claim, context, evidence organised by theme, the best counter-argument, and a recommendation. The course adds one requirement: at least one section reports a measurement you ran (a small audit from E2–E6, a replication of a published finding on a current model, or a survey of 30 classmates). Models change every quarter, so a replication of a 2023–2024 finding on 2026 models is itself a contribution.

Compare. your claim against its strongest alternative, using the literature plus your own measurement; for the replication option, the original paper's effect against the same probe on two 2026 models.
Measure. effect size and confidence interval of your probe; coverage of the literature (how many of the key papers you engage with); whether the recommendation is actionable (who, what, by when).
Deliver. the paper, the probe's code and data, and a one-page "if I am wrong" section listing what evidence would change your mind.
Week 1: claim, counter-claim, reading plan; Week 2: probe design and data collection; Week 3: analysis; draft argument; Week 4: revise against the counter-argument; poster. Start from: ICML call for position papers · Sorensen et al. and Kapoor et al. as model position papers · International AI Safety Report 2026 [85] for the state of the evidence · the writing tips.

E2. Do all languages cost the same? A dialect and minority-language audit of 2026 models Ahia et al. (EMNLP 2023) showed that speakers of many languages pay more tokens for worse results; Hofmann et al. (Nature 2024) showed that models judge people more harshly when they write in African American English. Replicate both designs for languages close to you: 粤语, 闽南语, 客家话, 藏语, 维吾尔语, 壮语, or a Southeast Asian language, against Mandarin and English. Use matched-guise probing: the same content, two language varieties, and decisions (job, credit, medical urgency) that should not change.

Compare. four models (two API, two open) on matched content across varieties; a tokenizer-level intervention (byte-fallback, vocabulary extension) or a prompt-level one (translate-then-answer) as the fix condition.
Measure. tokens per sentence, latency and price per task, task accuracy, and the decision gap between varieties with confidence intervals; agreement of two native-speaker raters on answer quality.
Deliver. the parallel probe set with native-speaker validation, the audit scripts, and a cost-and-quality

table per variety.
Start from: Ahia et al. · matched-guise probing code · Petrov et al., NeurIPS 2023 [86] (tokenizers introduce unfairness) · AceGPT (our lab: vocabulary expansion and native alignment for Arabic) · IrokoBench [87] as a format reference · pairs with B7.

E3. Matched-persona audits: gender and disability in generated decisions and recommendations Generate the texts that shape people's lives (reference letters, resume screens, performance reviews, loan explanations, career advice, triage notes) for personas that differ only in gender or in a disability cue, in Chinese and English. Wan et al. found "warm" language for women and "competent" language for men in LLM-written letters; Glazko et al. found resumes with a disability-related award ranked lower. Check whether 2026 models still do this, whether the pattern differs by language, and whether a mitigation (a rubric in the prompt, a critique pass) removes it without flattening everything.

Compare. matched personas across three models and two languages; baseline prompt vs. a rubric-guided prompt vs. a critique-and-revise pass.
Measure. lexical measures (warmth/competence, agentic/communal), ranking gaps, length and hedging differences, with bootstrap confidence intervals; a blinded human rating of a sample.
Deliver. the persona templates, the generated corpus, the analysis notebook, and a mitigation recommendation with its side effects.
Start from: Wan et al. · Glazko et al. · Gadiraju et al. · Blodgett et al. (define the harm before you measure it) · pairs with D8 for the rubric.

E4. Who gains? Novices vs. experts with an AI assistant on a real task, and what remains when it is removed Brynjolfsson, Li and Raymond (QJE 2025) found that customer-support novices gained most from an assistant; Bastani et al. (PNAS 2025) found that high-school students who practised with unrestricted GPT-4 scored lower when it was taken away, while a tutor-style version with guardrails did not hurt. Run a small version in your own cohort: one task (debugging, data analysis, a translation, a proof), two assistant designs, and a follow-up without the assistant.

Compare. no assistant · unrestricted assistant · a guardrailed assistant that explains but does not solve, between participants matched on a pre-test; a delayed post-test without any assistant.
Measure. task score and time with the assistant, post-test score without it, the novice–expert gap in both, self-reported reliance; keep n small (20–40) and report confidence intervals honestly.
Deliver. the protocol (pre-registered in your proposal), anonymised data, the analysis, and a note on what you would need to make the result reliable.
Start from: Generative AI at Work · Bastani et al. [23] (data) · Noy & Zhang · StudentBench [59] · follow the ethics guidance in the FAQ.

E5. An accessibility intervention, evaluated with the people it is for The benefit side of the disability question: multimodal models now describe images, read documents aloud, transcribe atypical speech and gloss sign language. Build one narrow tool with an accessibility organisation or a classmate who needs it, and evaluate it on tasks the users choose, not on a generic benchmark. Candidates: image descriptions for campus notices and lecture slides; a reading assistant for dyslexic students; speech recognition tuned to one dysarthric speaker; a Chinese Sign Language glossing aid; a plain-language rewriter for government forms.

Compare. the off-the-shelf model · your adapted or prompted version · the user's current workaround, on 20–30 user-chosen tasks.
Measure. task completion and time, error types that matter to the user (a missed warning is not the same as a missed adjective), satisfaction, and cost; with consent and anonymity throughout.
Deliver. the tool, the user-chosen task set, the evaluation protocol, and a short account of what the users said that the metrics missed.
Start from: how users with disabilities use chatbots (ASSETS 2024) · Whisper and Qwen3-VL · posts on X: Guided Vision in Gemini Live, ESP32 smart glasses from Kano, Nigeria, Visionauta · pairs with C19.

E6. Whose knowledge? A Global South audit with a retrieval intervention Ask practical questions (a fever in a child, a crop disease, a tenancy dispute, a scholarship deadline) as a user in Nairobi, Dhaka, Lima, Lagos and Lanzhou, and compare the answers with those for a user in San Francisco:

which sources are cited, which institutions are assumed, how often the advice is wrong or unusable locally. Then test the smallest fix: retrieval over local sources (ministry guidelines, local-language news, regional NGO documents) in front of the same model.

Compare. the bare model · model + retrieval over local sources · model + retrieval over global sources, for each location, same questions and model.
Measure. local correctness and usability rated by people from each place, source provenance (country of origin of cited sources), refusal and hedging rates, cost.
Deliver. the question set, local source corpora with licenses, rater protocol, and a provenance map of who the model listens to.
Start from: Sambasivan et al. · GlobalOpinionQA [84] · Santurkar et al. · the low-resource double bind · pairs with D14 for the evidence corpus.

E.4 Watch, read, play

- **Position-paper venues and examples** ICML position papers (since 2024) · NeurIPS position papers (since 2025) · ACL theme tracks · FAccT, CHI and ASSETS for audits with people · Science, Nature and PNAS for the economics and the large experiments Read two accepted position papers before writing one; note how each states its claim in the title.
- **Economics of AI, current** Canaries in the Coal Mine (Aug. 2026 revision: early-career employment in exposed jobs about 19% below trend, experienced workers flat) · Anthropic Economic Index · Humlum & Vestergaard · International AI Safety Report 2026 [85]
- **Accessibility in 2026 (X)** Google's Guided Vision for blind and low-vision users · AI smart glasses built on an ESP32 in Kano · Visionauta, by a blind engineer · Seeing AI Design prompts only; not independently verified.
- **Course and lab material** TA Rao Siqi's research direction is responsible AI and human value alignment; TA Liang Juhao works on multilingual models · AceGPT (Arabic localisation, NAACL 2024 and NeurIPS 2024) · Lecture 8, Topic 6: adapt an LLM to a new language

Pitfalls we see every year. A position paper with no position. An audit with ten prompts and no confidence intervals. "Bias" measured without saying what harm it causes to whom (read Blodgett et al. first). A human study with sensitive data and no consent. Claims about "the Global South" tested only in English. State the claim, measure it, report the interval, and say who should act.

F Efficiency, serving, on-device & GPU kernels

Inference is where most of the compute in the world now goes, and it is a track where **a single GPU and a careful measurement protocol produce publishable numbers**. Projects here study a systems trade-off: kernel generation with feedback, scheduling short and long requests together, quantization, speculative decoding, or edge deployment. Fix the workload and the measurement protocol before comparing methods; report latency distributions, not averages; and check correctness on fresh inputs, because a fast wrong kernel is the most common result in this area (the KernelBench-Verified paper is required reading on why reported speedups often evaporate). Two additions this term: an **on-device protocol** for phones and laptops (brief F8), and an **evolutionary, RSI-style loop** that optimises your own kernel or configuration under a sealed test (F9). F6 now includes the extreme case of a memory budget: a 2.78-trillion-parameter mixture-of-experts model streamed from disk on a single CPU.

Measurement protocol. Warm up, synchronize, repeat; report median and 95th percentile; pin the GPU clock or report its state; log the driver, CUDA and library versions; compare against both eager PyTorch and a compiled baseline where supported. Record power with `nvidia-smi` if you report energy. Keep the replay script in the repository.

F.1 Research questions this track asks

- On a fixed KernelBench subset, does iterative repair with compiler, correctness or profiler feedback beat independent candidates at a shared generation budget, and how many results are both correct and faster than each baseline?
- Which batched-token budget meets a stated time-to-first-token target at each request rate for a mixed short/long workload?
- Under identical prompts, how do serving engines differ in throughput, p50/p95 latency and energy per token?
- What accuracy does quantization or distillation cost on the tasks you care about, and where does it break first?
- How does speculative decoding's acceptance rate, and therefore its speedup, vary across domains and draft models?
- Can a real-time pipeline run on a laptop or phone with a bounded memory budget, and what is dropped to get there?
- When a mixture-of-experts model is far larger than memory, which split between resident layers and an expert cache, and which prefetch policy, buys the most tokens per second per gigabyte of RAM?
- On a phone, which runtime, quantization and drafting setting meets a stated latency target, and what does each cost in accuracy and energy?
- When an evolutionary agent optimises a kernel or a serving configuration, how much of the measured gain survives a sealed held-out test, and what did it try to game?

F.2 Featured briefs

F1. Feedback for GPU kernel generation on KernelBench KernelBench asks a model to replace a PyTorch reference module with a faster kernel: level 1 single operators, level 2 fused operators, level 3 full architectures. Pick a fixed subset and compare independent candidates with iterative repair that receives compiler errors, correctness failures on fresh inputs, or profiler traces, under a shared generation budget.

Compare. independent sampling · repair with error feedback · repair with profiler feedback, same budget.
Measure. correctness on fresh inputs, compilation time (separately), speedup against eager and compiled PyTorch, fraction of tasks both correct and faster than each baseline.
Deliver. the harness, generation traces, and the per-task table.
Week 1: install, reproduce one level-1 task end to end; Week 2: feedback loop variants; Week 3: run the subset, fresh-input checks; Week 4: profiles, table, write-up. Start from: KernelBench · KernelBench v0.1 · KernelBench-Verified [88] · TritonBench · GPU MODE lectures · for Ascend NPUs: CANN Bench [89].

F2. Serving short and long requests together Replay a mixed workload (short interactive requests, long prompts) on one model and one GPU. Vary the serving engine's batched-token budget and other scheduler knobs, and measure what each setting does to the two populations.

Compare. scheduler settings on the same arrival trace.
Measure. median and p95 time to first token, inter-token latency, throughput, by request class.
Deliver. a replay script and a note on which settings meet a stated latency target at each request rate.
Start from: vLLM optimization guide · SGLang · Lecture 11: Efficiency in LLMs.

F3. Engine comparison under identical prompts, with energy vLLM, SGLang, llama.cpp (and TensorRT-LLM if available) on the same model, prompts and hardware. The contribution is a careful protocol and an honest table, including joules per token.

Compare. engines and their default vs. tuned settings.
Measure. tokens per second, latency distribution, power and energy per token, output equivalence.
Deliver. bench harness, profiles, tuning guidelines.
Start from: vLLM · SGLang · llama.cpp.

F4. What quantization and distillation cost Quantize one model with two or three backends (AWQ, GPTQ, FP8) and optionally distil a smaller student; measure accuracy deltas on tasks that matter (long-context, math, code) rather than only perplexity. Pruning variants: LLM-Pruner, ShortGPT.

Compare. precision levels and backends at matched memory.
Measure. accuracy deltas per task, memory, throughput, where it breaks first.
Deliver. recipes and a break-point table.
Start from: llm-compressor · LLM-Pruner · ShortGPT [90].

F5. Speculative decoding across domains Measure acceptance rate and wall-clock speedup of speculative decoding (EAGLE-style or n-gram drafts) on code, chat, and long-form writing, with the same target model.

Bonus question: does the draft model have to be trained for each target ("teacher") model? EAGLE-style heads are trained on one target's hidden states, and a small draft is usually just a smaller sibling of the target. Take one draft and three targets (the original, a fine-tuned or RL-tuned version of it, and a different family with a compatible tokenizer) and measure how much acceptance falls. Then re-align the draft to each target, by distillation on the target's outputs (DistillSpec) or by online updates on live traffic (online speculative decoding), and report how many target-generated tokens of training buy back how much acceptance. Methods for heterogeneous vocabularies remove the shared-tokenizer constraint.

Compare. draft strategies and draft sizes; for the bonus, an off-the-shelf draft vs. a draft re-aligned to each target.
Measure. acceptance rate, speedup, output equivalence, memory.
Deliver. a benchmark script and a per-domain table.
Start from: EAGLE (and EAGLE-3 [19]) · vLLM speculative decoding · for the bonus: DistillSpec [20] · online speculative decoding [91] · speculative decoding with heterogeneous vocabularies [92].

F6. An edge real-time pipeline with limited memory Run a small model (ASR + LLM, or a VLM) on a laptop or phone under a stated memory budget and latency target. The research question is what to drop (precision, context, model size) and what it costs.

Bonus: how far can a memory budget stretch? kimi-k3-in-c runs the 2.78-trillion-parameter Kimi K3 on one CPU in portable C99, with no GPU and no BLAS. Only 16 of 896 experts fire per layer (about 104B active parameters), so the dense trunk stays in RAM while the 1.45 TB of routed 4-bit experts stream from an NVMe drive through an LRU cache with prefetching. The output is byte-identical from 8 GB of RAM (about 26 s per token) to 128 GB (about 6 s); only the clock changes, and its own measurements show that allocation beats capacity (at 128 GB, giving the trunk memory before the expert cache was worth 1.69×). Reproduce the idea on a mixture-of-experts model that fits your disk (OLMoE-1B-7B, Qwen3-30B-A3B), then test one policy of your own: frequency-based

caching, prefetching the next layer's experts from the router, a different trunk/cache split, or lower-precision experts, and plot tokens per second against gigabytes of RAM.

Compare. configurations under the same budget; for the bonus, LRU caching vs. your cache or prefetch policy at three or four memory budgets, on the same prompts, checking that the output stays identical.
Measure. end-to-end latency, memory, accuracy on a fixed task set, battery or power if possible; for the bonus, resident cache hit rate, the share of wall-clock spent on disk reads, and tokens per second per GB of RAM.
Deliver. a runnable app and the trade-off table.
Start from: llama.cpp · MLX · Qwen3-VL-2B/4B · for the bonus: kimi-k3-in-c [21] (Kimi K3 inference in C99 on one CPU, 8.24 GB peak RAM; Apache-2.0; its README documents every measurement) · Kimi K3 weights (1.56 TB: check your disk first) · smaller mixture-of-experts models: OLMoE, Qwen3-30B-A3B.

F7. Prefix-cache reuse for agent workloads Agent loops resend long, mostly identical contexts. Measure how much prefix caching and context layout (stable system prompt first, tool results last) save in an agent trace replay, and what it does to latency under concurrency.

Compare. caching on/off and two context layouts on the same replayed traces.
Measure. cache hit rate, prefill tokens, time to first token, throughput.
Deliver. the replay and a layout recommendation.
Start from: vLLM automatic prefix caching · context engineering for agents.

F8. An on-device benchmark: measure, then improve, a small model on a phone or laptop Gemma 4 E2B (2.3B effective, audio and vision input, Apache-2.0) runs on an iPhone 17 Pro at roughly 40 tokens per second with MLX; Qwen3-0.6B/1.7B, SmolLM3 and Liquid's LFM2.5 are the other usual suspects. The research question is a protocol, not a demo: on **your** device, which runtime, quantization, context length and speculative-decoding setting meets a stated latency target, and what does each setting cost in accuracy and battery? A fixed task set (50 prompts: QA, summarisation, a tool call, one image) and a fixed measurement script make the numbers comparable across teams.

Compare. runtimes (llama.cpp, MLC-LLM, ExecuTorch, MNN-LLM, MLX, Android AICore) · quantization levels (Q8, Q4, 2-bit) · with and without speculative or n-gram drafting, on the same model and prompts.
Measure. time to first token, decode tokens per second, peak memory, energy per 100 tokens (battery API or a USB power meter), accuracy on the task set, thermal throttling over a 10-minute run.
Deliver. a measurement script anyone can run on their phone, the trade-off table, and a recommended configuration per latency target.
Week 1: task set, measurement script, one runtime; Week 2: other runtimes and quantizations; Week 3: speculative decoding, energy, throttling; Week 4: table, recommendation, write-up. Start from: llama.cpp · MLC-LLM · ExecuTorch · MNN-LLM · MLX · Gemma 4 in Android AICore · Gemma 4 E2B/E4B · posts on X: Gemma 4 on an iPhone at 40 tok/s, M5 Max community benchmarks vs. the "4×" claim, LFM2.5-2.6B, an agentic on-device model, the local LLM playbook (Raspberry Pi to RTX 5090) · mllm-NPU for NPU prefill.

F9. Let an evolutionary agent (RSI-style) optimise your kernel or serving config, under a sealed test DeepMind's AlphaEvolve (May 2025) and its open-source counterpart OpenEvolve run a loop: an LLM proposes code edits, an evaluator scores them, a database keeps the best diverse programs, and the loop continues for hours. OpenEvolve's Metal-kernel experiment reported a 12.5% attention speedup on Apple silicon; Sakana's "AI CUDA Engineer" is the cautionary tale, where the loop found a way to cheat the benchmark. Apply the loop to one efficiency target you own, such as a Triton kernel from F1, a llama.cpp or vLLM configuration from F2/F3, or the on-device settings from F8, and measure whether the gains are real on inputs the loop never saw.

Compare. expert-tuned baseline · independent LLM proposals (same number of evaluations) · the evolutionary loop, with the evaluator seeing only the development inputs; a sealed held-out set and fresh random inputs for the final score.
Measure. speedup on development vs. sealed inputs (the overfitting gap), correctness failures, evaluations to the first real improvement, GPU hours spent by the loop itself, and at least one reward-hacking attempt found in the traces.

Deliver. the loop configuration, the program database, the sealed-test script, and the best program with a human-readable explanation of why it is faster.
Week 1: target, evaluator, sealed set, expert baseline; Week 2: run the loop; inspect traces; Week 3: independent-proposal control; fresh-input checks; Week 4: explain the winner, write-up. Start from: OpenEvolve · OpenEvolve GPU-kernel discovery post · AlphaEvolve · AI CUDA Engineer and its reward-hacking postscript · ShinkaEvolve · RSI-Exam (brief A5) · autonomous nanoGPT speedrun agents · TA Ke Ji's research direction is recursive self-improvement; ask in office hours.

F.3 Watch, read, play

- Video: **Stanford CS25 V6: Serving Transformers, Lessons from the Trenches**. What production serving teams measure, and why.
- **Live competitions in this area** FlagOS / IEEE: optimise LLM operators and inference across AI chips (listed as closing mid-November; verify) · AMD + GPU MODE: optimise LLM inference on MI355X (listed on ML Contests; verify dates) · public KernelBench CUDA and KernelBench Mega leaderboards See Track G for the full list with dates.
- **2026 reading on kernel agents** KernelBench-Verified [88] · KernelArc [93] · FastKernels [94] · KernelGenBench [95] Common thread: reward hacking and measurement noise; your fresh-input checks are the defence.
- **On-device numbers (X)** Gemma 4 E2B on iPhone 17 Pro, about 40 tok/s with MLX · M5 Max: dense 27B at 23 tok/s, MoE 122B at 65 tok/s · LFM2.5-2.6B runs tools on-device · "watch the tokens per second: that number is your reality check" Numbers quoted from posts; reproduce them with the F8 protocol before citing them.
- **Evolutionary optimisation and RSI** OpenEvolve · ShinkaEvolve · AI CUDA Engineer (read the reward-hacking addendum) · RSI-Exam · auto-nanoGPT (Codex and Claude Code iterating for two weeks) The sealed test is the whole method; without it the loop optimises your evaluator.
- **Course material** Lecture 11: Efficiency in LLMs · GPU MODE lecture series · vLLM docs

Pitfalls we see every year. Timing without warm-up or synchronization. Correctness checked on the inputs the model saw. A speedup against an unoptimized baseline only. Averages instead of distributions. A thermal-throttled GPU in one condition. Keep the protocol in the repo and run both conditions back to back, interleaved.

G Competitions: a formal final-project option

A competition project uses a contest task, dataset or simulator to answer a course research question. You may enter an open competition or run an archived task locally; **both count as final projects** and use the course poster and report requirements. The leaderboard is a bonus result, not the grade: what we assess is the controlled comparison you ran on a local held-out split. A contest whose final deadline falls after Dec. 25 is fine, as long as the course deliverables are on time.

1. **Register a course plan.** Identify the contest, track, task, official metric, and the change you will test. Note the model, data, runtime and submission rules (offline notebooks, allowed models, API bans).
2. **Run a controlled comparison.** Reproduce one baseline, keep a local held-out split or fixed simulator scenarios, and compare your method at a matched budget. Use the public leaderboard for an additional result when available.
3. **Submit course deliverables.** Report local results, resource use and error analysis; include code and instructions to reproduce the baseline and the proposed method; add a dated competition score or submission link if you entered.

Dates below were checked on November 3, 2026. Unless marked AOE (Anywhere on Earth, UTC–12), Kaggle deadlines are 23:59 UTC. Always confirm on the organizer's page before planning around a date. open entry open now · closing soon within two weeks · verify listed by aggregators, date not confirmed · archived run locally.

G.1 Chinese competitions (中文比赛)

- **知乎知学堂复杂表格识别与理解 Tianchi · listing**. *Dates:* open through Dec. 31, 2026; *Task and a research change to test:* Recover complex tables from images and PDFs; automatically scored Excel answers. Compare a vision-language model with an OCR-plus-structure pipeline; report merged cells, value extraction and arithmetic separately.; *Track:* F, multimodal
- **Qoder 高校创意大赛 rules**. *Dates:* open register and post by Nov. 30, 2026, 18:00 Beijing; *Task and a research change to test:* University teams of 1–3 build an application with Qoder, deploy it on Qoder Sites, and post a write-up. For a StudyHack-style entry, compare a document assistant with and without evidence citations on a fixed set of research tasks.; *Track:* F, C
- **OpenLET 具身智能挑战赛 competition and schedule**. *Dates:* open winter simulation round Sep. 29 –Dec. 31, 2026; *Task and a research change to test:* Simulation-based manipulation with a provided policy. Compare the frozen policy with a small adaptation or a recovery rule; report success by placement and perturbation (brief C7). Hardware rounds come later.; *Track:* B
- **CCKS 2026 评测任务 (知识图谱与语义计算大会) CCKS 2026 · hosted on Tianchi**. *Dates:* verify per task; *Task and a research change to test:* Tasks include a cultural-tourism knowledge graph (entity extraction and disambiguation, cross-modal knowledge expansion) and an LLM behaviour-control evaluation. Even after deadlines, the task definitions and data are useful local benchmarks.; *Track:* F
- **NLPCC 2026 shared tasks official list**. *Dates:* archived evaluation closed June–Sep. 2026; conference Nov. 2026, Macau; *Task and a research change to test:* Public data and metrics: Task 6 three-way LLM-text detection (brief D3); Task 10 reliability of AI-assisted scientific reporting (D4); Task 1 [96] medical instructional video QA; Task 9 AI Scientist Benchmark.; *Track:* F
- **多模态客服 Agent 设计 (DataFountain) archived specification**. *Dates:* archived ran Mar. 16 –Aug. 31, 2026; *Task and a research change to test:* A text-and-image chat endpoint backed by a knowledge base. Recreate the task with documents and images you may use; compare text-only and image-aware retrieval; measure grounded accuracy and appropriate abstention.; *Track:* F
- **蚂蚁灵波具身大模型挑战赛 Tianchi**. *Dates:* archived online round closed Oct. 26, 2026; *Task and a research change to test:* Embodied VLA tasks; the released task format is a ready-made local benchmark for Track C manipulation studies.; *Track:* B
- **CCF BDCI 2026 (大数据与计算智能大赛) 息壤 platform**. *Dates:* archived registration closed Sep. 10, 2026; *Task and a research change to test:* Three Agent tasks (for example,

quantitative investment report generation and automatic research-paper generation on the JiuwenSwarm framework). Task specifications define evaluation you can reuse.; *Track:* A

- **讯飞 AI 开发者大赛智能体赛道 / 星火杯 agent track · 星火杯**. *Dates:* verify last agent cycle Dec. 26, 2025 –Apr. 9, 2026; *Task and a research change to test:* Campus agent-building contests on the 星辰 Agent platform; the previous cycle opened in late December, so a new cycle may open during the project window.; *Track:* A, C
- **中国高校计算机大赛 C14-AI 2026 (鸿蒙 / 昇腾 tracks) official site**. *Dates:* archived 2026 finals announced; next edition spring 2027; *Task and a research change to test:* Teams of up to three; AI application projects on HarmonyOS or Ascend. A Track C or C prototype built this term can be carried into the 2027 edition.; *Track:* any

G.2 Kaggle, NeurIPS 2026 and other international competitions

- **Google DeepMind: Gemma 4 Developer Agent competition · model update · paper track**. *Dates:* open paper track Nov. 12; entry Nov. 25; final Dec. 2, 2026; *Task and a research change to test:* Post-train an open model into a coding agent that repairs real repository issues; held-out tests score patches offline with an approved Gemma 4 variant (31B, 12B, E4B, E2B). Compare repository search alone with search plus a test-and-repair loop (brief A7).; *Track:* A
- **BuildArena S02: Human–AI Engineering Challenge tracks and rules**. *Dates:* closing soon Nov. 14, 2026 AOE (see Kaggle countdown); *Task and a research change to test:* Agent-only design vs. natural-language copilot guidance in a physics simulator. Only for teams with a working prototype; deliver a controller, telemetry and an interaction transcript (brief A3).; *Track:* A
- **ARC Prize 2026 Paper Track Kaggle · arcprize.org**. *Dates:* closing soon papers due Nov. 8 (arcprize.org) / Nov. 9 (Kaggle), 2026; *Task and a research change to test:* A write-up documenting a real ARC-AGI-2 or ARC-AGI-3 submission; the prediction tracks closed Nov. 2. The public tasks remain available for program-search and memory studies (brief A8).; *Track:* A
- **Name That LLM competition and rules**. *Dates:* open entry Jan. 6; final Jan. 13, 2027; *Task and a research change to test:* Identify which of 18 models generated a passage; offline notebooks, nine-hour limit; public data and pretrained models allowed. Compare character n-grams with a small encoder; report confusion across model families (brief D3).; *Track:* F
- **Pokémon TCG AI Battle Challenge Playground competition**. *Dates:* open entry Jan. 1; final Jan. 8, 2027; *Task and a research change to test:* Use the simulator SDK to compare a heuristic policy with one that searches over legal actions; report win rate against fixed opponents and time per move. Verify execution restrictions before designing around a hosted model.; *Track:* A
- **NeurIPS 2026 Smart Buildings Challenge Kaggle · site**. *Dates:* open entry Jan. 3; final Jan. 15, 2027; *Task and a research change to test:* Compare a fixed HVAC controller with a learned or model-assisted policy in simulation; measure comfort, energy cost and emissions on held-out scenarios. Offline notebooks, nine-hour limit.; *Track:* A
- **FlagOS / IEEE: optimise LLM operators and inference across AI chips ML Contests listing**. *Dates:* verify listed as closing Nov. 15, 2026; *Task and a research change to test:* Operator and inference optimization across hardware; pairs with brief F1. Confirm the date and rules on the organizer page.; *Track:* H
- **AMD + GPU MODE: LLM inference on MI355X ML Contests listing**. *Dates:* verify; *Task and a research change to test:* Inference performance optimization on AMD GPUs with a large prize pool; requires access to the hardware through the organizer. Protocol lessons apply to brief F3.; *Track:* H
- **Google Quantum AI: optimise quantum circuits (Kaggle) Kaggle competitions**. *Dates:* verify about three months to go; *Task and a research change to test:* Tagged agents / code / optimisation on ML Contests; an agentic program-optimization problem for teams comfortable with the domain.; *Track:* A
- **NeurIPS 2026 competition track all 16 competitions**. *Dates:* verify most close in November; track meets Dec. 11–12; *Task and a research change to test:* Relevant entries: Steerability Challenge, Predictive AI Evaluation, QuantiPhy (VLM physical reasoning), SimulacraBench, AIMO Interpretability, RoboSynChallenge, Agenthon (verifiable AI for finance), Speech Accessibility Project Challenge 2.; *Track:* F, D, B

G.3 Archived tasks and benchmarks you can run locally

- **ARC-AGI-2 and ARC-AGI-3**. *What it gives you:* Public grid tasks and interactive environments; the 2026 prediction tracks closed Nov. 2, winners announced Dec. 4.; *Track:* A
- **AIMO Progress Prize 3**. *What it gives you:* 50 public olympiad-style problems selected to be "AI-hard"; closed Apr. 15, 2026. A hard set for test-time scaling studies.; *Track:* A, D
- **KernelBench** · **TritonBench**. *What it gives you:* 250 kernel tasks in three levels; public leaderboards for CUDA and megakernel variants.; *Track:* H
- **GameCraft-Bench [our lab]**. *What it gives you:* 140 Godot game tasks with a replay-based verifier and rubric judge.; *Track:* A
- **RSI-Exam**. *What it gives you:* 88 sealed self-improvement tasks across six domains; contributions for 2.0 listed for Dec. 15, 2026.; *Track:* A
- **SWE-bench** · **Terminal-Bench** · **OSWorld**. *What it gives you:* Repository repair, terminal tasks and computer-use tasks with executable checks.; *Track:* A
- **AgentBench** · **VisualAgentBench**. *What it gives you:* Multi-environment agent evaluation from Tsinghua; function-calling variant based on AgentRL.; *Track:* A
- **LIBERO** · **ManiSkill**. *What it gives you:* Manipulation benchmarks with released VLA checkpoints (SmolVLA-LIBERO on LeRobot).; *Track:* B
- **LongBench** · **AlignBench**. *What it gives you:* Long-context and Chinese alignment evaluation suites.; *Track:* F
- **AgentDojo**. *What it gives you:* Prompt-injection attacks and defenses for tool-using agents.; *Track:* F
- **ML Contests** · **Kaggle** · **Tianchi** · **DataFountain** · **讯飞** · **Codabench**. *What it gives you:* Where to look for contests that open after this page was written.; *Track:* any

Rules that bite. Many Kaggle tasks score notebooks offline with no internet and a nine-hour limit: an API-model pipeline cannot be submitted, though it can still be your local baseline. Read the allowed-models clause before choosing a brief. Chinese contests often require real-name registration and platform accounts (Tianchi, 息壤, 星辰); start that in week 1.

H AI entrepreneurship: run a startup prototype

Lecture 8's last topic invites you to run a prototype for a startup: an **AI CEO**, an **AI seller**, an **AI tutor**, **AI for expertise training** (simulated patients for medical students, interview coaching), an **AI game designer** (text-to-game, text-to-app), an **open-world experience** with AI and VR, or **speech-based 3D printing**. Two student companies from this lab show the scale that is possible: 薯塔 AI, an AI study platform for Sino-foreign universities, had covered 60% of CUHK-Shenzhen undergraduates by April 2025 (3,579 registered on campus, 4,680 in total, 1,239 daily active users and about 2,900 AI answers a day) and is expanding to seven other universities; 网事, a tennis app by 朱宇轩, has served over a thousand players and saved them more than 5,000 hours. A startup project is still a course project: it needs real users, a controlled comparison, and a report. What changes is the evidence you collect: activation, retention, willingness to pay, and a business memo that survives a sceptical reader.

What counts as a research result here. Two versions of your product on the same user tasks (onboarding A vs. B, retrieval over course materials vs. a generic model, free vs. paid tier), with the metric decided in advance. Ten user interviews written up with the Mom Test rules (ask about their life, not your idea). A unit-economics table: cost per active user per month (API tokens, hosting, support) against what they pay or would pay. A one-page business memo. The poster shows the comparison and the retention curve, not the logo.

H.1 AI startup basics (what every founder in this track should be able to explain)

Problem before product. Write the customer's problem in their words before touching a model. Interview ten people with the Mom Test rules: past behaviour, not opinions about your idea. The lab's successful teams started from a course they were failing and a sport they played.

Business model canvas. Customer segments, value proposition, channels, revenue, cost structure, key activities and partners. The canvas or the Lean Canvas fits on one page; fill it in week 1 and again in week 4 and show what changed.

AI unit economics. Cost per active user per month = tokens $\times$ price + hosting + human review. Compare with ARPU (average revenue per user) and CAC (acquisition cost). Gross margins of AI apps are lower than classic software because inference is a cost of goods; caching, small models and routing are product decisions.

The "wrapper" question. If the model vendor ships your feature tomorrow, what remains? Proprietary data, a workflow embedded in a customer's day, distribution (a campus, a hospital, a WeChat group), trust and brand, switching costs. Name your moat in one sentence or admit you have none yet.

Product-market fit signals. Retention curves that flatten rather than decay to zero; organic referrals; users complaining when it breaks; the Sean Ellis test ("very disappointed" if it disappeared $\geq$ 40%). Vanity metrics: sign-ups, demo views, likes.

Pricing. Per seat, per usage, per outcome, freemium. AI products are moving toward outcome and usage pricing because costs scale with use. Test one price with a real "buy" button even if you refund everyone; stated willingness to pay is not a measurement.

Compliance in China and abroad. Generative AI services facing the public in China fall under the 生成式人工智能服务管理暂行办法 (2023) and algorithm filing rules; personal data under PIPL; the EU AI Act for European users. Education and health products have extra rules. Write the one paragraph that says what applies to you.

Build fast, measure faster. Vibe-coding tools make a prototype a weekend's work; the scarce resource is users and the discipline to measure. Ship to 20 users in week 2, not 2,000 in week 4. Read Paul Graham's "Do things that don't scale" and the YC library.

H.2 Research questions this track asks

- Does retrieval over a course's own materials beat a general model for a student tutor on correctness and on the student's willingness to return next week?
- Can an AI seller raise conversion in a real small shop without raising refund or complaint rates, and which messages it sends make the difference?
- When an agent runs a business end to end, which decisions does it get wrong first (pricing, inventory, promises to customers), and does a rule layer fix them without killing revenue?
- Do simulated patients or interviewers improve a trainee's performance with real humans, or only their comfort with the simulator?
- From a sentence to a playable game or a printable object: what fraction of prompts yields something usable, and what does a repair loop add per attempt?
- Which onboarding, pricing or guardrail variant keeps more users after two weeks, at what cost per retained user?

H.3 Featured briefs

H1. AI tutor for one course, with retention as the metric Build a tutor for one real course (ask the instructor), grounded in its slides, notes and past assignments, and give it to classmates. 薯塔 AI grew from exactly this. The product variable is grounding; the business variable is whether students come back. Design the tutor so it explains rather than solves (Bastani et al., PNAS 2025) and measure whether that costs you users. Do not start from zero: HKU's open-source DeepTutor already combines citation-grounded answers, difficulty-calibrated practice questions and a learner memory, and ships TutorBench for evaluation; use it as your codebase or as the baseline to beat.

Compare. a generic chat model · the same model with retrieval over the course corpus · the grounded tutor with a "no final answers on graded work" guardrail, assigned to matched groups of students.
Measure. answer correctness on 50 course questions (graded by the TA), week-2 retention, questions per active user, cost per active user, a learning check on a quiz without the tutor.
Deliver. the deployed tutor, the retention table, ten user interviews, a one-page unit-economics memo.
Week 1: course corpus, interviews, baseline deployed to 20 users; Week 2: grounded and guardrailed variants; Week 3: retention and learning measures; Week 4: memo, poster. Start from: DeepTutor (HKU Data Intelligence Lab; paper [22]) · C17 for memory design · StudentBench [59] · Bastani et al. [23] · posts on X: Brilliant's Koji, Aristotle refuses to give answers, a $6M seed round for a tutor that makes kids figure it out.

H2. AI seller: a sales and support agent for a real small shop Find one small business (a campus café, a family's online store, a tutoring centre) and build the agent that answers product questions, recommends, and closes orders in its chat channel, with a human approving anything irreversible. The research question is what the agent may say: a free conversational policy against one constrained by an approved product sheet and a refund policy.

Compare. human-only (baseline week) · agent with free policy · agent with a constrained policy and escalation rules, on the same channel in alternating weeks or A/B by customer.
Measure. conversion, average order, response time, complaints and refunds, hallucinated promises caught by the human reviewer, cost per conversation.
Deliver. the agent, the escalation rules, the week-by-week table, and the shop owner's verdict in their own words.
Start from: A4 for the harness · Dify or LangGraph · τ^2-bench for retail-style agent evaluation before going live.

H3. AI CEO: an agent runs a tiny business, and you measure where it fails Anthropic's Project Vend let Claude run a small office shop for a month: it found suppliers and answered customers, but it also sold at a loss, gave discounts to anyone who asked, invented a supplier contact and had an identity crisis. Andon Labs' Vending-Bench turns the same setting into a simulator. Run an agent as the manager of a tiny real or simulated business (a vending box in your dorm, a print-on-demand shop, a weekly snack pre-order) and study its decisions.

Compare. the bare agent · agent + a rule layer (price floors, inventory limits, a human sign-off on promises) · agent + a daily self-review memo, over the same simulated weeks or alternating real

weeks.
Measure. profit, stock-outs, pricing errors, promises it could not keep, customer satisfaction, and a timeline of the first serious mistake under each condition.
Deliver. the agent, the ledger, replays of the failures, and a memo on which decisions an AI manager should never make alone.
Start from: Project Vend (thread) · Vending-Bench · TheAgentCompany · the "startup simulator" idea from Lecture 8.

H4. AI for expertise training: simulated patients, interview coaches, exam examiners Medical students practise on simulated patients; job seekers practise interviews; law students practise moots. The lab's AI Hospital and Agentic Hospital papers simulate patients and examiners with LLMs. Build the trainer for one skill with a partner who teaches it (a clinical teacher, a career centre, a debate coach), and measure transfer to a human assessor, not just scores inside the simulator.

Compare. no practice · practice with a scripted simulator · practice with the LLM simulator plus rubric feedback, followed by an assessment with a real person using the same rubric.
Measure. rubric score with the human assessor, number of practice sessions, trainee confidence, simulator realism rated by the teacher, cost per session.
Deliver. the simulator with persona library, the rubric, the transfer results, and the teacher's assessment of where the simulator misleads.
Start from: AI Hospital [97] and Agentic Hospital (our lab) · HuatuoGPT [43] · D8 for rubric design · Lecture 9.

H5. AI game designer: text-to-game and text-to-app as a product The lab's GameCraft-Bench measures whether agents can turn a spec into a playable Godot game; vibe-coding jams show solo builders shipping multiplayer games in a weekend. The product question is the user, not the model: a teacher who wants a quiz game, a parent who wants a story game for a child, a small brand that wants a promotional mini-game in WeChat. Build the generator with a repair loop and test it with ten such users.

Compare. one-shot generation · generation + automatic replay-and-repair · generation + repair + one round of user edits in plain language, on the same 30 user prompts.
Measure. launchable rate, playable rate judged by the user, time to an accepted game, cost per accepted game, and what users asked to change.
Deliver. the generator, 30 games, the user-edit log, and a pricing experiment with at least one real payment.
Start from: A2 and GameCraft-Bench · Claude-Code-Game-Studios · posts on X: a capybara delivery game, solo in 14 days, a Hong Kong jump game from text prompts, 20 hours, 500 prompts, 20 euros.

H6. Open-world experience: an AI + VR product people pay for A VR experience generated or narrated by AI: a guided tour of a historical site with a character who answers questions, a language-immersion café, a calming space that adapts to the visitor. Build on C9 (WebXR assistant) and C2 (generated assets), then run sessions with real visitors and ask what they would pay for.

Compare. scripted experience · AI character with a static briefing · AI character with memory and retrieval, with visitors assigned to one condition.
Measure. session length, questions asked, a recall quiz afterwards, motion sickness and comfort, willingness to pay (a real price test), cost per session.
Deliver. the experience on Quest or WebXR, session logs, the visitor survey, and a memo on cost per session against price.
Start from: DecartXR · LLMR [53] · HY-World 2.0 for scenes.

H7. Speech-based 3D printing: from a spoken wish to a printed object "A phone stand that also holds my earbuds." The pipeline is speech recognition, a CAD-code generator (parametric CadQuery or build123d, or Blender Python as in the lab's BlenderLLM), a printability check (manifold, overhangs, dimensions), and a slicer. Open-source text-to-CAD agents appeared in 2026 for Claude Code and Codex; Adam CAD sells speech-to-3D. Your product variable is the repair loop and the questions the system asks before it prints.

Compare. one-shot generation · generation + printability repair loop · the loop plus one clarifying question about dimensions or use, on 30 spoken requests from 10 users.

Measure. printable rate (manifold, fits the bed), dimension error against the user's stated numbers, user acceptance of the printed object, filament and time per accepted object.
Deliver. the pipeline, 30 STL files with provenance, photos of printed objects, and the acceptance table.
Start from: BlenderLLM [98] (our lab; model) · Text2CAD (NeurIPS 2024) · CadQuery / build123d · PrusaSlicer CLI · posts on X: text-to-cad in the Codex plugin directory, a /render skill with build123d, an open-source CAD engineer for Claude Code and Codex, Adam CAD speech-to-3D · pairs with A3.

H.4 Watch, read, play

- Video: **Andrej Karpathy, "Software Is Changing (Again)"**, YC AI Startup School, June 2025. Software 3.0, partial autonomy and why "vibe coding" is a product strategy question.
- **Founder reading** YC Startup Library and Requests for Startups · Do things that don't scale · The Mom Test · Lean Canvas · Sequoia: AI's $600B question · a16z on the AI app layer · 奇绩创坛 (MiraclePlus) for a Chinese accelerator's view
- **Lab startups and brainstorm slides** 薯塔 AI (AI study platform; 60% of CUHK-Shenzhen undergraduates by April 2025) · 网事 (tennis app, 1,000+ users) · Lecture 8 brainstorm slides: robot bar, AI seller, expertise training, open-world VR, text for 3D printing, text2game, a startup simulator Teams with a working prototype can ask the instructor about incubation contacts and the CUHK-Shenzhen hospital deployment track (email wangbenyou@cuhk.edu.cn).
- **Product demos on X** Project Vend · text-to-cad · a vibe-coded game that shipped · a vibe-coded weekend project on Hacker News · "show me one vibe-coded game worth playing" Both the enthusiasm and the scepticism are data about your market.

Pitfalls we see every year. A pitch deck instead of a measurement. "Users" who are the team's friends and never return. A chatbot wrapper with no moat and no cost table. Promises to customers that an agent made and nobody caught. Collecting personal data without consent. Ship to twenty real users, measure retention and cost, and write the memo a sceptical investor would want.

I Learn from other courses and labs

Several courses run projects very much like ours. Their public project archives, assignment chains and final-project lectures are worth an evening each: they show what a four- to ten-week team can finish, and what reviewers reward. Links were checked on Nov. 3, 2026.

Stanford CS224N: NLP with Deep Learning (Diyi Yang, Winter 2026).

- Project page with award-winning projects and links to every past report archive (2015–2024).
- 2026 lecture on choosing a final project and the report instructions.
- Recent outstanding projects to study: FlashAttention / Longformer / cosFormer for GPT-2; KV caching and speculative decoding at GPT-2 scale; TRACE (tool-augmented reasoning via atomic cheatsheet editing); dynamic token merging for encoder–decoders.
- Their default project ("efficient adaptation of GPT-2 via LoRA") is a template for a bounded Track B study.

Stanford CS336: Language Modeling from Scratch (Liang, Hashimoto; Spring 2026).

- Course site; Spring 2026 lectures on YouTube; 中英字幕 on Bilibili; assignment repos.
- Five assignments: tokenizer + Transformer + optimizer from scratch; Triton FlashAttention and hand-rolled DDP; scaling laws from small runs; data curation from Common Crawl; SFT, DPO and reasoning RL.
- Maps to briefs B1, B2, B10, D12 and F1. Their advice applies here: disable AI autocomplete while learning the core pieces, then use agents for the scaffolding.

Stanford CS329X: Human-Centered LLMs (Diyi Yang, Fall 2026).

- Course site with public slides (human preferences, RLHF/DPO crash course, design thinking).
- Project categories: apply or extend an LLM to a real problem; new ways to learn from human feedback; fairness and bias; interactive systems; persona simulation. Top projects have become conference papers.
- Class-wide paper: Reflections and New Directions for Human-Centered LLMs [56] (60+ student authors). Maps to Track C and brief C16.

Stanford CS329Z: Engineering AI Agents (Diyi Yang, Michael Ryan, John Yang; Fall 2026).

- Syllabus, slides and homework: RAG, tool use, agent loops, DSPy and LangGraph, data curation, evaluation; final demos Dec. 7–11.
- Structured around three challenges: decomposition, data, evaluation. Lecture 6 (design patterns and scaffolds) and lecture 8 (multi-agent systems) are the Track A readings.
- HW1 builds a company assistant with no framework, "just a chat-completion call and code you write yourself" (brief A4). Chinese study site.

Stanford CS25: Transformers United V6 (Spring 2026).

- Course site · YouTube playlist (V1–V6) · Bilibili (V1–V4).
- V6 talks used on this page: world modeling (Track C), the ultra-scale training talk (B), serving transformers (F), native multimodal intelligence, collaborative science agents.

Tsinghua 高级机器学习 (Advanced Machine Learning) (唐杰 Tang Jie).

- 2024 archive (original pages on aminer.cn): HW1 reproduce GLM-4-9B basics; HW2 fine-tuning and RAG; HW3 visual LLM understanding and generation; HW4 advanced reasoning. Team projects in reasoning, multimodal models, recommendation, education, science, healthcare and business.
- Fall 2026 edition (as reported): 16 weeks from base models through scaling, synthetic data, preference and RL, agents, continual learning; assignments from a hand-written tokenizer and Transformer to a 0.1B model, a Triton attention kernel, scaling-law fitting, a

LoRA/QLoRA/adapter comparison, and a long-horizon self-evaluating agent; teams of 2–3, NeurIPS-format English report, live demo in week 16.
- Open-source projects from the group and Z.ai to build on: GLM-5.x, slime (asynchronous RL), Open-AutoGLM (phone agent), AgentBench, CogVideoX, CogAgent (GUI agent VLM), LongBench, AlignBench, GLM-V.

Our lab: FreedomIntelligence [CUHK-Shenzhen].

- GameCraft-Bench (playable-game generation benchmark, brief A2) · Agentic Hospital and AI Hospital [97] (hospital as an agent environment, brief H4).
- Other lab work used by briefs on this page: Soundwave (speech, B12), AceGPT (Arabic localisation, B7/E2), BlenderLLM (CAD, H7), ORLM [46], RAG-Instruct [48], Happy-Medical-AI (C19), Huatuo-26M (D13), MileBench and MLLM-Bench (D7).
- Medical LLMs: HuatuoGPT [43], HuatuoGPT-II [44]; see Lecture 9.
- Teams who want to extend a lab benchmark or environment can ask the TAs for a scoped sub-task during office hours.

J Toolbox & media

A curated set of models, libraries, simulators and compute options that the briefs above assume. Prefer tools with permissive licenses and active maintenance; record the exact version you used.

Frontier and open models.

- API: Claude Opus 5.5 and other frontier models as planner, coder, judge; log model and role.
- Open weights: GLM-5.x (MIT for most releases), Qwen3.x, DeepSeek, Gemma 4.
- Small models for one GPU: Qwen3-0.6B/1.7B/4B, Gemma 4 E2B/E4B, SmolLM, the Tutorial 2 bases.

Vision, speech, video, 3D.

- VLMs: Qwen3-VL (2B–235B, Apache-2.0), GLM-4.6V, InternVL3.5, DeepSeek-OCR for documents.
- Speech: Whisper, CosyVoice 3, Qwen3-TTS, IndexTTS.
- Video: Wan 2.2, HunyuanVideo 1.5, LTX-2, CogVideoX.
- 3D and worlds: TRELLIS.2, Hunyuan3D 2.1, HY-World 2.0, interactive world models list.
- Dance and motion: EDGE, AtomicDance, AIST++.

Web 3D, games, lessons.

- three.js (glTF loading, TransformControls), three-vrm, Mixamo.
- Godot 4 for GameCraft-Bench; Playwright for isolated browser runs and screenshots.
- Mol*, GlowScript, Manim, Remotion, Motion Canvas.
- Prototyping UIs: Gradio, Streamlit.

Agents and harnesses.

- Claude Code, Codex CLI, OpenClaw, Cline, Roo Code.
- DSPy, LangGraph, MCP, A2A.
- Benchmarks: SWE-agent, Terminal-Bench, OSWorld, AgentBench, ToolBench.

Training and data.

- nanochat, CS336 assignments, TRL, PEFT.
- RL: verl, slime, OpenRLHF, SimpleRL.
- Data: datatrove, FineWeb-Edu, Dolma.
- Interpretability: Gemma Scope, SAELens, TransformerLens.

Serving and kernels.

- vLLM, SGLang, llama.cpp, MLX.
- llm-compressor, EAGLE.
- KernelBench, TritonBench, GPU MODE lectures, Triton.

Evaluation and safety.

- lm-evaluation-harness, Inspect AI.
- LongBench, AlignBench, AgentDojo, ALCE.

Simulators and robots.

- AI2-THOR, Habitat 3.0, ManiSkill, LIBERO.
- MuJoCo, Genesis, LeRobot, LLaVA-VLA.

On-device and XR.

- MLC-LLM, ExecuTorch, MNN-LLM, Android AICore (Gemma 4), Gemma 4 E2B/E4B.
- WebXR, A-Frame, Immersive Web Emulator, LLMR [53].
- World models: V-JEPA 2, jepa-wms, DreamDojo, DreamerV3.

Compute.

- Kaggle: free T4/P100 sessions (about 30 GPU hours per week); the Tutorial 2 notebooks run there.
- Colab; AutoDL (hourly GPUs in China); Hugging Face Spaces for demos; Modal for burst jobs.
- Ask the teaching team early if a brief needs more than one GPU for days; plan the small-scale version first.
- **Huawei Ascend**: Lecture 8 announced possible access to Ascend NPUs for training and inference adaptation; teams interested in porting a brief to CANN should say so in the proposal.

Course material to revisit.

- Lecture 7: Agents and harness engineering · Lecture 8: Final projects (Fall 2026 deck) · Writing tips, poster and paper samples.
- Lecture 9: Medical LLMs · Lecture 10: Multimodal LLMs · Lecture 11: Efficiency.
- Tutorial 1 (PyTorch, embeddings) · Tutorial 2 (SFT, LoRA/QLoRA, PPO, GRPO, DPO) · AI tools for research.